ROBSTRIDE DYNAMICS



ROBSTRIDE 03

60N.M Quasi-Direct Drive Integrated Motor Module

User Manual

RobStride Dynamics

https://www.robstride.com/

RS03 instruction manual

Precautions

1. Please use according to the working parameters specified in this article, otherwise it may cause serious damage to the product!

2. Do not switch the control mode when the joint is running. If you need to switch, send the command to stop the operation before switching.

3. Check whether the parts are in good condition before use. If the parts are missing or damaged, contact technical support in time.

4. Do not disassemble the motor at will, so as to avoid unrecoverable failure.

5. Ensure that there is no short circuit when the motor is connected, and the interface is correctly connected as required.

Legal Statement

Before using this product, please read this manual carefully and operate the product according to the contents of this manual. If the user violates the contents of this manual to use this product, resulting in any property damage, personal injury accident, the company does not assume any responsibility. Because this product is composed of many parts, do not allow children to touch this product to avoid accidents. In order to prolong the service life of the product, do not use this product in high temperature and high pressure environment. This manual has been printed to the extent possible to include a description of the functions and instructions for use. However, due to the continuous improvement of product functions, design changes, etc., there may still be discrepancies with the products purchased by users.

The color and appearance of this manual may differ from the actual product. Please refer to the actual product. This manual is published by Beijing Lingfoot

Times Technology Co., LTD. (hereinafter referred to as Lingfoot), and Lingfoot may at any time make necessary improvements and changes to the inaccurate and up-to-date information in this manual, or make improvements to procedures and/or equipment. Such changes will be uploaded to the company's official website in electronic format. Details can be found in the download center (www.robstride.com). All images are for reference only. Please refer to actual objects.

After-sales Policy

The after-sales service of this product is implemented in strict accordance with the Law of the People's Republic of China on the Protection of Consumer Rights and Interests and the Product Quality Law of the People's Republic of China. The service content is as follows:

1. Warranty period and contents

    a. Users who place orders on the online channel to purchase this product can enjoy the return service without reason within seven days from the day after signing. When returning goods, the user must present a valid proof of purchase and return the invoice. The user must ensure that the returned goods maintain the original quality and function, the appearance is intact, the trademarks and various logos of the goods themselves and accessories are complete, and if there are gifts, they should be returned together. If the goods are artificially damaged, artificially disassembled, missing packaging boxes, missing parts and accessories, they will not be returned. The logistics cost incurred during the return shall be borne by the user (see "After-sales Service Fee Standard"). If the user does not settle the logistics cost, it will be deducted from the refund amount according to the actual amount incurred. Refund the amount paid to the user within seven days from the date of receipt of the returned item. Refund method is the same as payment method. The

specific arrival date may be affected by factors such as banks and payment institutions.

b. The warranty period of this product is 1 year.

c. Within 7 days after the user signs for the next day, non-human damage performance failure occurs, through the Lingzhu after-sales service center test and confirmation, for the user to handle the return business, the user must present a valid purchase voucher, and return the invoice. Any freebies should be returned.

d. From 7 days to 15 days after the user signs for the next day, non-human damage performance failure occurs, through the Lingfoot after-sales service center test and confirmation, for the user to replace the whole set of goods. After the replacement, the three guarantee period of the goods themselves is recalculated.

e. From 15 days to 365 days after the user signed the next day, after the inspection and confirmation of the Lingfoot after-sales service center, it is a quality fault of the product itself, and can provide free maintenance services. The replacement of the faulty product is owned by Lingzu Company. The product is not faulty and will be returned as is. This product has been strictly tested after the factory, if there is a quality fault other than the product itself, we will have the right to refuse the user's return demand.

2. Non-warranty regulations The following circumstances are not covered by the warranty:

3. Exceed the warranty period specified in the warranty terms.

4. Failure to follow the instructions, resulting in product damage caused by wrong use.

5. Damage caused by improper operation, maintenance, installation, modification, testing and other improper use.

6. Non-quality failure caused by conventional mechanical loss, wear.

7. Damage caused by abnormal working conditions, including but not limited to falling, impact, liquid immersion, violent impact, etc.

8. Damage caused by natural disasters (such as floods, fires, lightning strikes, earthquakes, etc.) or incapacitated forces.

9. Damage caused by exceeding peak torque.

10. Damage caused by exceeding peak torque.

11. Failure or damage caused by other non-product design, technology, manufacturing, quality and other problems.

12. Use this product for commercial purposes.

In the case of the above situation, the user must pay the cost.

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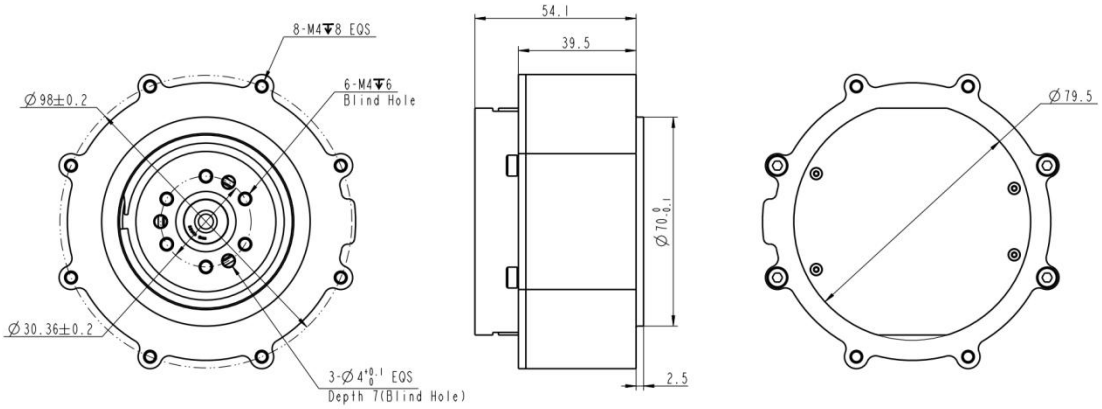
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1. Motor specification

1.1. Outline and mounting dimensions



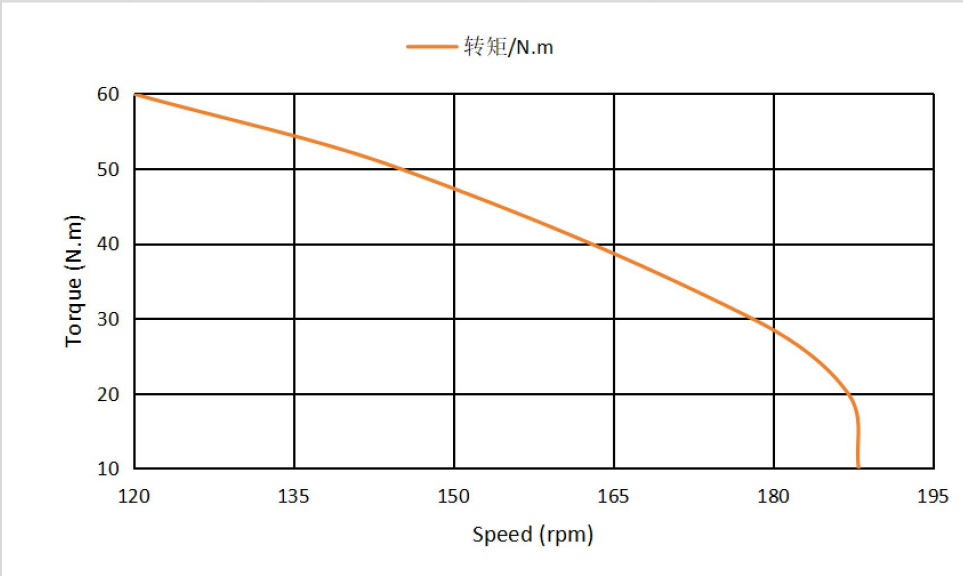
When fixing, the screw depth should not exceed the depth of the casing thread

1.2. Standard service condition

1. Rated voltage: 48 VDC
2. Operating voltage range: 24V-60 VDC
3. Rated load (CW) : 20 N.m(Heat sink dimensions: 215mm × 220mm)
4. Operation direction: CW/CCW from the direction of the exit shaft
5. Use posture: the direction of the exit axis is horizontal or vertical
6. Standard operating temperature: 25±5℃
7. Operating temperature range: -20 ~ 50℃
8. Standard operating humidity: 65%
9. Humidity range: 5 ~ 85%, no condensation
10. Storage temperature range: -30 ~ 70℃
11. Insulation Class: Class B

1.3. Electrical characteristics

1. No load speed: 200 rpm±10%
2. No load current: 0.6 Arms
3. Rated load: 20 N.m(Tested at 100 rpm)
5. Rated load phase current (peak) : 13Apk±10%
6. Peak load: 60 N.m
7. Maximum load phase current (peak) : 43Apk±10%
8. Insulation resistance/stator winding: DC 500VAC, 100M Ohms
9. High voltage/stator and housing: 600 VAC, 1s, 2mA
10. Motor back potential: 17Vrms/krpm±10%
11. Torque constant(Valid value): 2.36N.m/Arms
12. T-N curve



13. Maximum overload curve

Test Conditions:

Ambient Temperature: 25℃

Heater Material: Aluminum Alloy

Dimensions: 215mm × 220mm

Winding Limit Temperature: 145℃ (This is the constraint temperature; the actual temperature is 180℃)

Protection Logic:

Speed: 6nm tested at 100rpm

Test Data(Reduced heat sink size, lower rated torque)

Loat	Operating time(s)
60	13
50	34
40	67
30	393
20	rated

14. Thermal testing Test Conditions:

The winding temperature of a single phase is calculated in real time using the motor phase current and the motor thermistor temperature sensor. When the motor enters a stalled state, the three phases heat unevenly. The system calculates the temperature of each phase based on the current, and triggers protection when the calculated temperature reaches the thermal threshold.

PS: Under stall conditions, the heat generation of a single phase is 1.414 times that under rotating conditions. The stall overload time is as follows.

Loat	Operating time(s)
60	1
50	4
40	9
30	25
20	190
13	rated

1.4. Mechanical characteristic

1. Weight: 880g±20g
2. Number of poles: 42
3. Phase number: 3 phases
4. Drive mode: FOC
5. Deceleration ratio: 9:1

2. Driver Product Information

2.1. Driver product Specifications

project	data
The rated working voltage	48VDC
The maximum allowable voltage	60VDC
Rated working phase current	12Apk
Maximum allowable phase current	43Apk
Standby power	≤40mA
CAN bus bit rate	1Mbps
Dimensions	Φ70mm
Working environment temperature	-20℃ to 50℃
The maximum allowable temperature of the control board	145℃
encoder resolution	14bit (absolute turn)

2.2. Driver interface definition



2.3. Driver interface recommended brand and model

board end model	brand manufacturer	line end model	brand manufacturer
XT30APW-M	AMASS (Ams)	XT30UW-F	AMASS (AMS)
GH1.25-2PWT	any	GH1.25-T	any

2.4. Driver function pin and device description

1. Power supply and CAN communication

Pin	description
1	The positive electrode of the power supply (+)
2	Negative electrode of the power supply (-)

Pin	description
3	CAN CAN_L
4	CAN the high side of the communication CAN_H

1. Download port

Pin	description
1	SWDIO (data)
2	SWCLK (clock)
3	3V3 (positive 3.3V)
4	GND

1. Indicator light

Driver function pin and device description

3. Upper computer instructions

Please go to www.robstride.com website download center

3.1. Hardware disposition

The articulated motor uses the CAN communication mode and has two communication cables. It is connected to the debugger through the can to USB tool. The debugger needs to be installed with the ch340 driver in advance and works in AT mode by default.

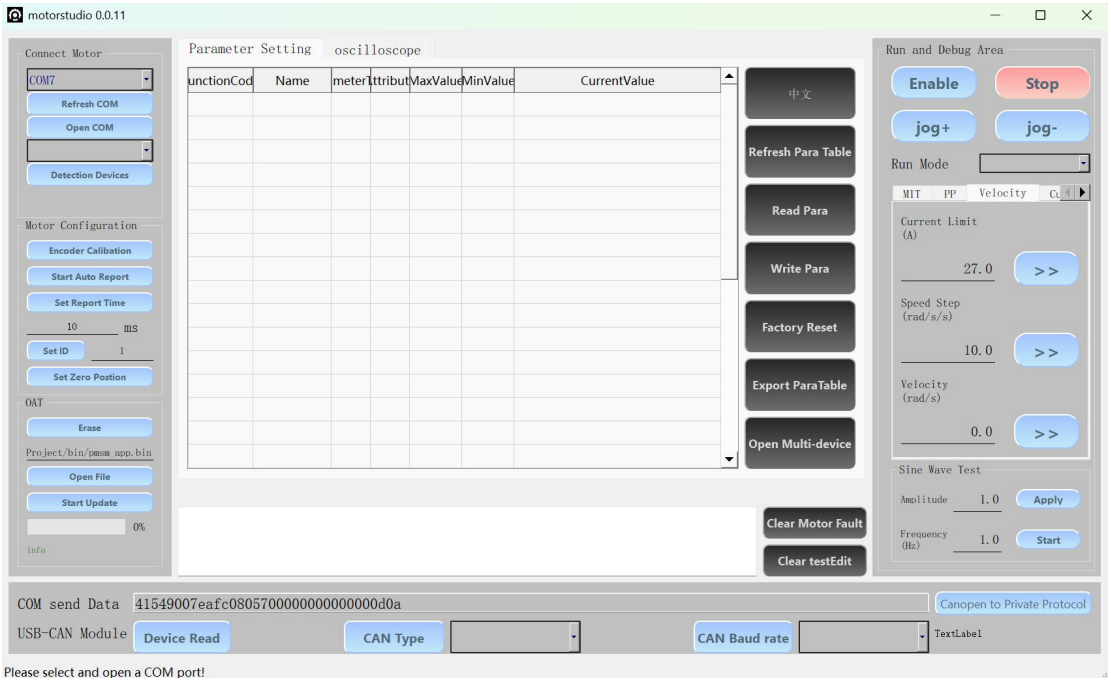
It should be noted that we are based on the specific can to USB tool development of the debugger, so we need to use our recommended serial port tool to debug the debugger, if you want to transplant to other debugger platform can refer to the third chapter of the instructions for development.

The CAN to USB tool is recommended to use the official USB-CAN module of Lingzu Times. The frame header of the corresponding serial port protocol is 41 54, and the frame tail is 0D 0A.

When using the CAN-to-USB module, pay attention to the settings of the DIP switches on the module: When DIP switch 1 is in the ON position, the module

enters Boot mode and cannot establish a connection with the host computer. When DIP switch 2 is in the ON position, a 120Ω terminal resistor is connected to the module port, allowing normal communication with the host computer.

3.2. Upper computer interface and description



Mainly includes:

A. Motor Connection Module

- Refreshing the Serial Port
- Opening the Serial Port
- Testing the Device

B. Motor Configuration Module

- Starting the Upgrade
- Opening a File
- Starting the Upgrade
- Modifying the Motor CAN ID
- Setting the Motor's Mechanical Zero Position

C. Motor Upgrade Module

- Magnetic Encoder Calibration
- Motor Active Reporting Switch
- Setting the Motor Active Reporting Time
- Modifying the Motor CAN ID
- Setting the Motor's Mechanical Zero Position

D. Motor Main Interface

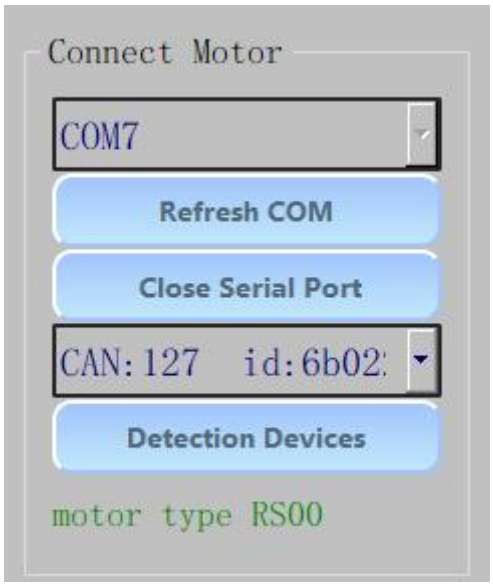
- Parameter Settings
- Motor Oscilloscope

E. Run and Debug Area

- Parameter Debugging Buttons
- Motor Mode Configuration and Parameter Modification
- Sine Signal Testing

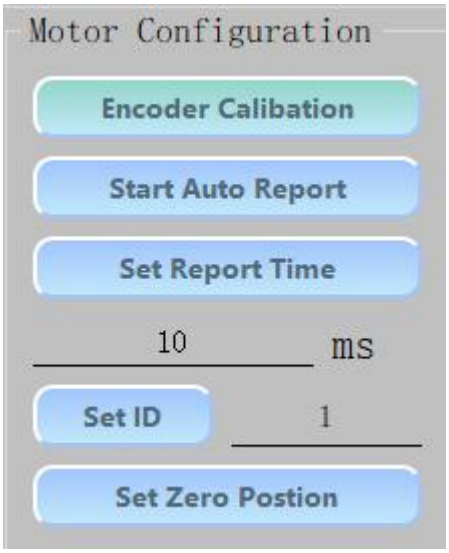
3.3. Motor settings

3.3.1. Motor connection settings



Connect the CAN to USB tool (install the ch340 driver, which works in AT mode by default), click Refresh Serial Port, open the serial port, and click Detect Device to detect the corresponding motor. The green text below is the motor type.

3.3.2. Motor configuration module



1. Recalibrate the motor magnetic encoder. Reinstalling the motor board and motor, or reconnecting the motor's three-phase wiring requires recalibrating the magnetic encoder.

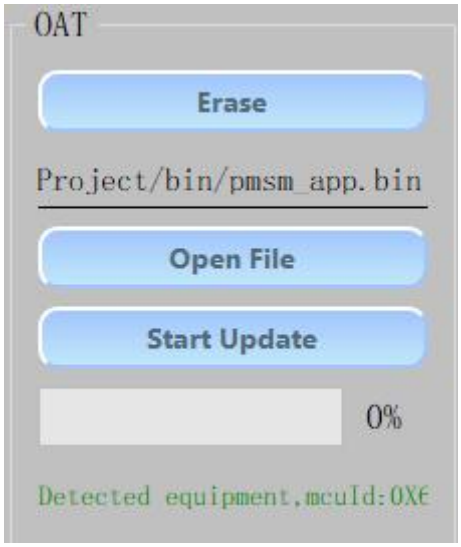
2. Enable active motor reporting. Click Start Reporting to enable active motor reporting in communication type 2. You can set the interval below, with a minimum of 10ms.

3. Set ID: Set the motor's CAN ID.

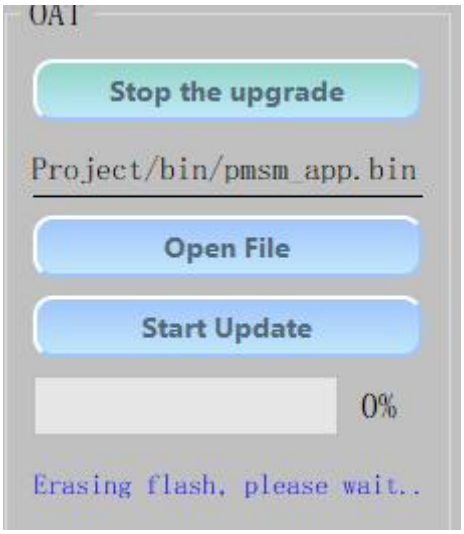
4. Set Zero Position: Set the current position to 0.

3.3.3. Motor upgrade module

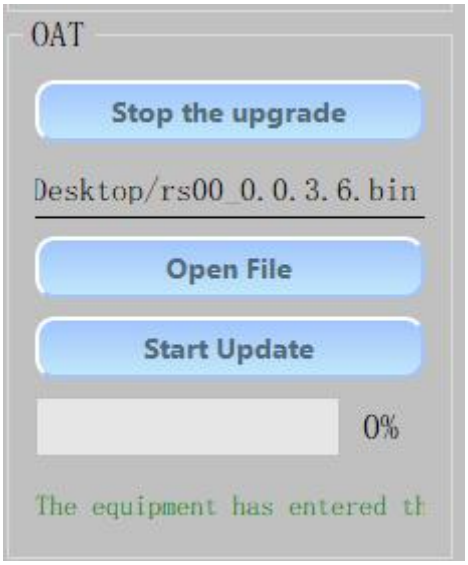
1. 1. Click to open the file and select the firmware to upgrade. The rs-0x in the firmware name is the selected motor type.



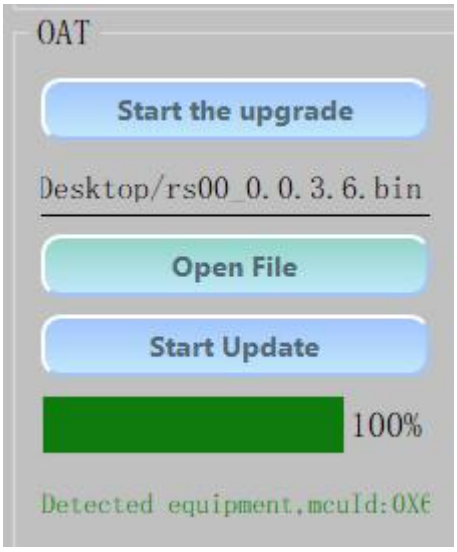
2. Click Start Upgrade, and the motor will enter the upgrade preparation stage.



3. When the green text "Device has entered upgrade mode" pops up, click to start the upgrade

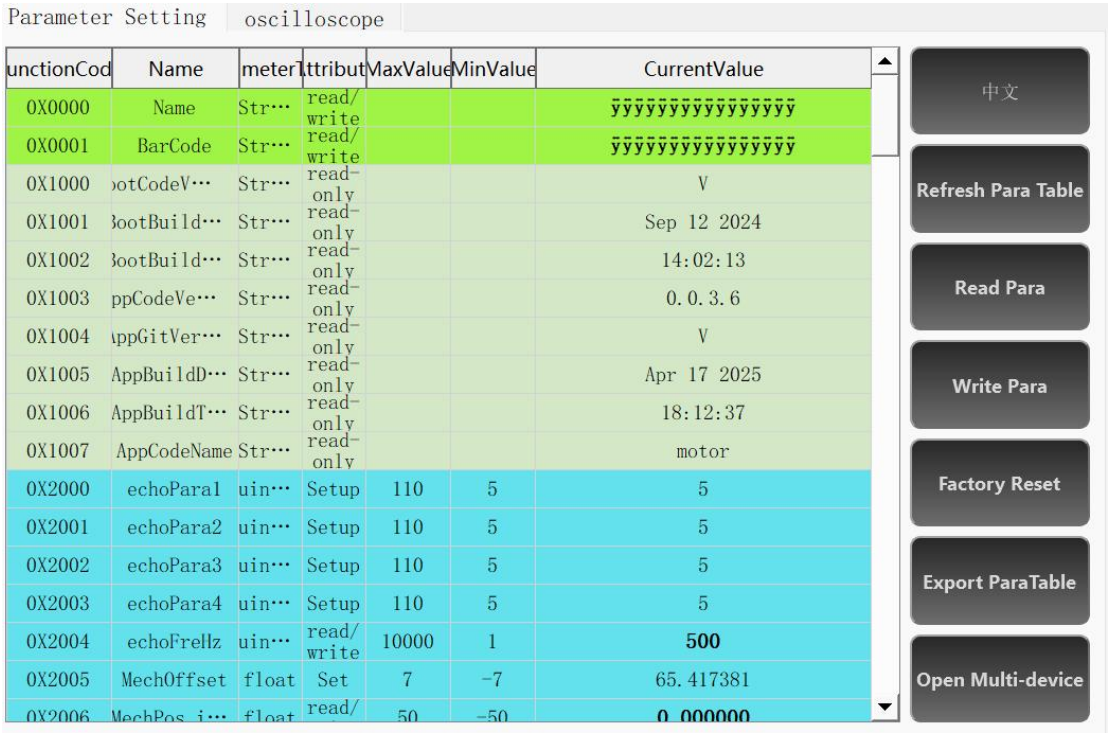


4. When the green text "Upgrade Successfully" pops up, the upgrade is complete.



If the green progress bar gets stuck halfway through the upgrade, you can click to stop the upgrade, or re-power on and re-enter the upgrade process. The internal program of the motor will not be lost after the upgrade fails. Please check whether the communication environment is good before upgrading again.

3.3.4. Parameter settings



After successfully connecting to the motor,

1. Click Refresh Parameter Table. "Updated Parameter Table Successfully" will appear at the top, indicating that the motor parameters have been successfully read (Note: The parameter table must be configured while the motor is in standby mode. If the motor is running, the parameter table refresh cannot be performed). The interface will display the motor's parameters. Parameters in blue are stored internally in the motor and can be modified in the Current Value field following the corresponding parameter.

2. Click Read Parameters to upload the motor parameters to the debugger. Parameters in light blue are observed parameters, which are collected and can be observed in real time.

3. Click Write Parameters to download the debugger parameters to the motor.

4. Click Restore Factory to restore the motor's default parameters for the latest firmware.

5. Click Export to export the current motor parameters in the parameter table.

6. Click Open Multi-Device Connection to connect the host computer to multiple motors. Note that because the parameter interfaces for different motor types vary, multi-device connection is only used for upgrades. After upgrading and debugging the motor, close multi-device connection and search for the motor again.

Note: Please do not change the torque limit, protection temperature and overtemperature time of the motor. Our company will not bear any legal responsibility for any damage to human body or irreversible damage to joints caused by illegal operation of this product.

function code	name	parameter type	attribute	Maximum value	Minimum value	Current value (for reference)	备注
0X0000	Name	String	Read/Write			ÿÿÿÿÿÿÿÿÿÿÿÿÿÿÿÿ	
0X0001	BarCode	String	Read/Write			ÿÿÿÿÿÿÿÿÿÿÿÿÿÿÿÿ	
0X1000	BootCodeVersion	String	Read only			0.1.5	
0X1001	BootBuildDate	String	Read only			Mar 16 2022	
0X1002	BootBuildTime	String	Read only			20:22:09	
0X1003	AppCodeVersion	String	Read only			0.0.0.1	Motor program version number
0X1004	AppGitVersion	String	Read only			7b844b0fM	
0X1005	AppBuildDate	String	Read only			Apr 14 2022	
0X1006	AppBuildTime	String	Read only			20:30:22	
0X1007	AppCodeName	String	Read only			Lingzu_motor	
0X2000	echoPara	uint16	dispositio	74	5	5	

function code	name	parameter type	attribute	Maximum value	Minimum value	Current value (for reference)	备注
	1		n				
0X2001	echoPara2	uint16	disposition	74	5	5	
0X2002	echoPara3	uint16	disposition	74	5	5	
0X2003	echoPara4	uint16	disposition	74	5	5	
0X2004	echoFreHz	uint32	Read/Write	10000	1	500	
0X2005	MechOffset	float	Settings	7	-7	4.619583	Motor magnetic encoder Angle offset
0X2006	Chasu_offset	float	Read/Write	50	-50	4.52	Reserved parameter
0X2007	Status1	float	Read/Write	17	0	17	Torque limitation
0X2008	I_FW_MAX	float	Read/Write	33	0	0	Weak magnetic current value, default 0
0X2009	CAN_ID	uint8	Settings	20	0	1	Motor index, marking the joint position of the motor
0X200a	CAN_MASTER	uint8	Settings	127	0	1	id of this object
0X200b	CAN_TIMEOUT	uint8	Settings	127	0	0	can host id
0X200c	status2	uint32	Read/Write	100000	0	0	can timeout threshold. The default value is 0
0X200d	status3	int16	Read/Write	1500	0	800	Reserved parameter
0X200e	Status4	uint32	Read/Write	1000000	1000	20000	Reserved parameter
0X200f	Status5	float	Read/Write	64	1	7.75	Reserved parameter
0X2010	Status6	uint8	Read/Write	1	0	1	Reserved parameter
0X2011	cur_filt_gain	float	Read/Write	1	0	0.9	Current filtering parameter
0X2012	cur_kp	float	Read/Write	200	0	0.025	Current kp
0X2013	cur_ki	float	Read/Wri	200	0	0.0258	Current ki

function code	name	parameter type	attribute	Maximum value	Minimum value	Current value (for reference)	备注
			te				
0X2014	spd_kp	float	Read/Write	200	0	2	Velocity kp
0X2015	spd_ki	float	Read/Write	200	0	0.021	Speed ki
0X2016	loc_kp	float	Read/Write	200	0	30	Position kp
0X2017	spd_filt_gain	float	Read/Write	1	0	0.1	Velocity filter parameter
0X2018	limit_spd	float	Read/Write	200	0	2	Location mode speed limit
0X2019	limit_cur	float	Read/Write	23	0	23	Position, Velocity mode current limit
0X201a	limit_a	float	Read/Write	100	0	0	Reserved parameter
0X201b	fault1	float	Read/Write	100	0	0	Reserved parameter
0X201c	fault2	float	Read/Write	27	0	0	High speed segment offset
0X201d	fault3	float	Read/Write	27	0	0	The low end is offset
0X201e	fault4	float	Read/Write	150	0		Velocity-mode acceleration
0X201f	fault5	float	Read/Write	20	0		PP mode speed
0X2020	fault6	float	Read/Write	1000	0		PP mode acceleration
0X2021	fault7	float	Read/Write	100	0	0	Zero marker
0X2022	baud	uint8	Read/Write	10	0	0	Baud rate flag
0X2023	zero_sta	uint8	Read/Write			0	Zero point flag
0X2024	position_offset	uint8	Read/Write	27	0	0	Position offset
0x2025	protocol_1	uint8	Read/Write			0	Protocol flag
0x2026	damper	uint8	Read/Write	0	20	0	Damping switch

function code	name	parameter type	attribute	Maximum value	Minimum value	Current value (for reference)	备注
0x2027	add_offset	float	Read/Write	-7	7	0	Position offset parameter
0X3000	timeUse0	uint16	Read only			5	
0X3001	timeUse1	uint16	Read only			0	
0X3002	timeUse2	uint16	Read only			10	
0X3003	timeUse3	uint16	Read only			0	
0X3004	encoderRaw	int16	Read only			11396	Magnetic encoder sampling value
0X3005	mcuTemp	int16	Read only			337	mcu internal temperature, *10
0X3006	motorTemp	int16	Read only			333	Motor ntc temperature, *10
0X3007	vBus(mv)	uint16	Read only			24195	Bus voltage
0X3008	adc1Offset	int32	Read only			2084	adc sampling channel 1 Zero current bias
0X3009	adc2Offset	int32	Read only			2084	adc sampling channel 2 Zero current bias
0X300a	adc1Raw	uint16	Read only			1232	adc sampling value 1
0X300b	adc2Raw	uint16	Read only			1212	adc sampling value 2
0X300c	VBUS	float	Read only			36	Bus voltage V
0X300d	cmdId	float	Read only			0	id ring instruction, A
0X300e	cmdIq	float	Read only			0	iq ring command, A
0X300f	cmdlocref	float	Read only			0	Position loop command, rad
0X3010	cmdspdref	float	Read only			0	Speed loop command, rad/s
0X3011	cmdTorque	float	Read only			0	Torque instruction, nm
0X3012	cmdPos	float	Read only			0	mit Protocol Angle instruction
0X3013	cmdVel	float	Read only			0	mit Protocol Speed instruction
0X3014	rotation	int16	Read only			1	Number of turns
0X3015	modPos	float	Read only			4.363409	Motor uncounted coil mechanical Angle, rad
0X3016	mechPos	float	Read only			0.777679	Load end loop mechanical Angle, rad
0X3017	mechVel	float	Read only			0.036618	Load speed: rad/s

function code	name	parameter type	attribute	Maximum value	Minimum value	Current value (for reference)	备注
0X3018	elecPos	float	Read only			4.714761	Electrical Angle
0X3019	ia	float	Read only			0	U-wire current, A
0X301a	ib	float	Read only			0	V-wire current, A
0X301b	ic	float	Read only			0	W-wire current, A
0X301c	timeout	uint32	Read only			31600	Timeout counter value
0X301d	phaseOrder	uint8	Read only			0	Directional marking
0X301e	iqf	float	Read only			0	iq filter value，A
0X301f	boardTemp	int16	Read only			359	Plate temperature，*10
0X3020	iq	float	Read only			0	iq Original value，A
0X3021	id	float	Read only			0	id Original value，A
0X3022	faultSta	uint32	Read only			0	Fault status value
0X3023	warnSta	uint32	Read only			0	Warning status value
0X3024	drv_fault	uint16	Read only			0	The driver chip fault value is 1
0X3025	drv_temp	int16	Read only			48	The driver chip fault value is 2
0X3026	Uq	float	Read only			0	Q-axis voltage
0X3027	as_angle	float	Read only			0	Magnetic encoder initial angle
0X3028	cs_angle	float	Read only			0	Differential magnetic encoder initial angle
0X3029	chasu_angle	float	Read only			0	Differential angle
0X302a	v_bus	float	Read only			0	Motor voltage
0X302b	ElecOffset	float	Read only			24.195	Electrical angle offset
0X302c	torque_fdb	float	Read only			0	Torque feedback value，nm
0X302d	rated_i	float	Read only			8	Rated current of motor
0X302e	MechPos_init	float	Read only			27	Motor retention parameters
0X302f	instep	float	Read only			0	Motor protection parameters
0X3030	status	uint8	Read only			0	Reserved parameters
0X3031	cmdlocre	float	Read only			0	Position setpoint

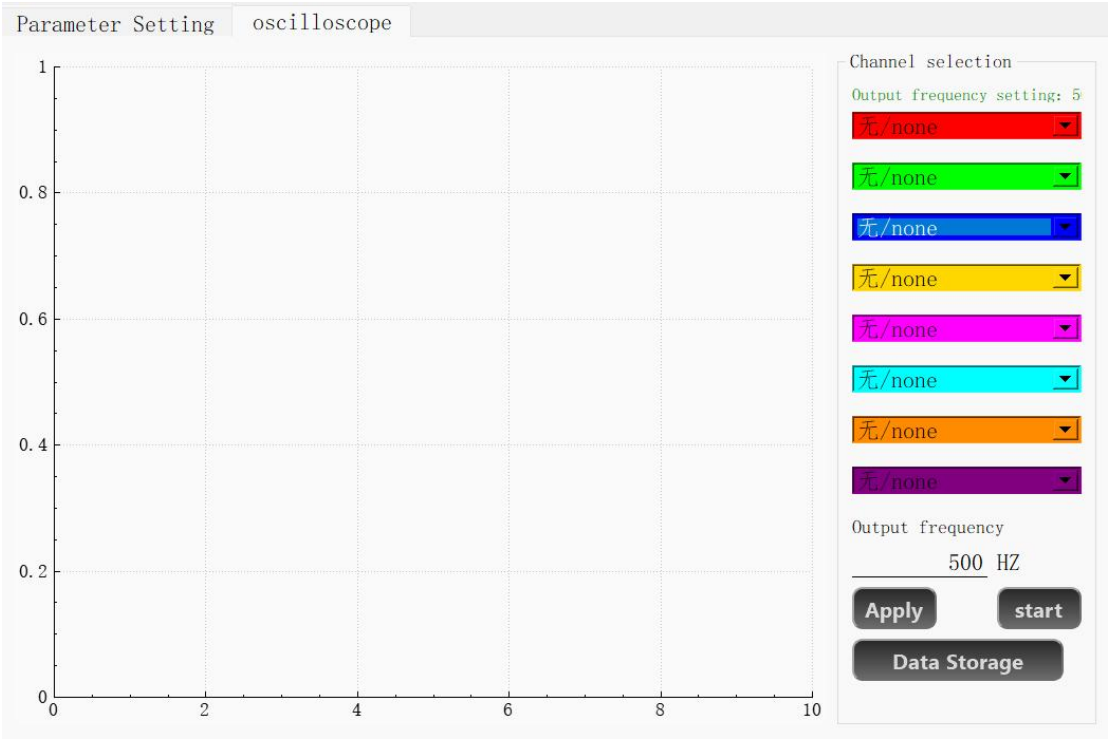
function code	name	parameter type	attribute	Maximum value	Minimum value	Current value (for reference)	备注
	f						
0X3032	vel_max	float	Read only			0	Motor speed setpoint
0X3033	fault1	float	Read only			0	Motor position determination parameters
0X3034	fault2	float	Read only			0	Motor position determination parameters
0X3035	fault3	float	Read only			0	Motor position determination parameters
0X3036	fault4	float	Read only			0	Motor position determination parameters
0X3037	fault5	float	Read only			0	Log failure
0X3038	fault6	uint32	Read only			0	Log failure
0X3039	fault7	uint32	Read only			0	Log failure
0X303a	fault8	uint32	Read only			0	Log failure
0X303b	mcOverTemp	int16	Read only			0	Over-temperature threshold
0X303c	Kt_Nm/Amp	float	Read only			0	Torque coefficient
0X303d	Tqcali_Type	uint8	Read only			0	Motor type
0X303e	theta_mech_1	float	Read only			0	Type 2 low-speed angle
0X303f	adc0ffset_1	uint32	Read only			2053	ADC sampling channel 1 zero current offset
0X3040	adc0ffset_2	uint32	Read only			2057	ADC sampling channel 2 zero current offset
0X3041	can_status	uint8	Read only			1	CAN status
0X3042	position	float	Read only			16.558939	Initialization position
0X3043	chasu_an	float	Read only			2.106539	Motor position

function code	name	parameter type	attribute	Maximum value	Minimum value	Current value (for reference)	备注
	gle_init						determination parameters
0X3044	chasu_angle_out	float	Read only			18.958853	Motor position determination parameters
0X3045	motormechinit	float	Read only			16.558939	Motor position determination parameters
0X3046	mech_angle_init2	float	Read only			1.839882	Motor position determination parameters
0X3047	mech_angle_rotat	int16	Read only			0	Motor position determination parameters
0X3048	coder_reg	uint16	Read only			1	System parameters
0X3049	pos_cnt1	uint16	Read only			0	System parameters

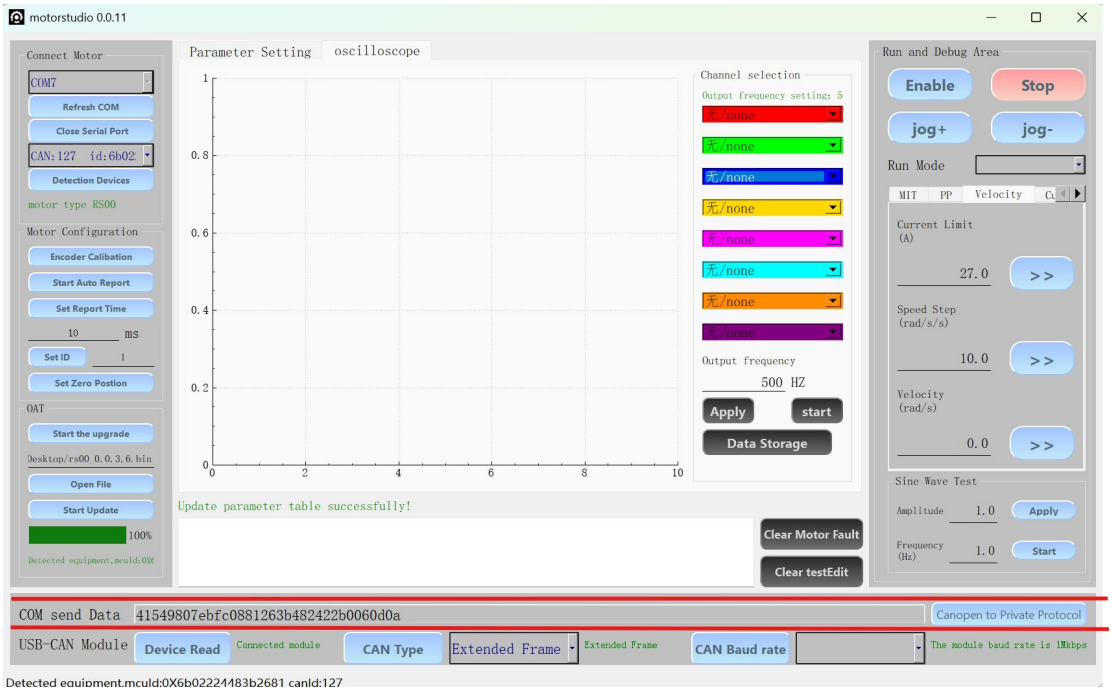
3.3.5. Oscilloscope

The interface supports viewing and observing the graph generated by real-time data, including motor Id/Iq current, temperature, real-time speed at the output end, rotor (encoder) position, output end position, etc.

Click on the oscilloscope module in the analysis module, select the appropriate parameters in the channel (parameter meaning can be referred to the parameter table), set the output frequency, click on the start plot to observe the data graph, stop the plot to stop the observation graph.



The command sent is in the communication command box below



Communication box instruction example:

41 54 90 07 e8 0c 08 05 70 00 00 01 00 00 00 0d 0a

The meaning is as follows

41 54	**90 07 e8 0c**	**8**	**05 70 00 00 01 00 00**	**0d 0a**

			00	
frame header	Number of data bits	extended frame	data frame	frame tail

The translation of extended frame canid into real canid requires the following transformations:

90 07 e8 0c converts to binary as 1001 0000 0000 0111 1110 1000 0000 1100, remove the 100 on the right and it becomes 1 0010 0000 0000 1111 1101 0000 0001, convert it to hexadecimal, It is 12 00 FD 01. According to the communication protocol, the meaning is as follows:

12 in hexadecimal	**0**	**FD**	**1**
Communication type 18 (in decimal base)	No meaning	host id	motor canid

3.3.6. can communication failure protection

When the value of CAN_TIMEOUT is 0, this function is disabled

When the CAN_TIMEOUT value is non-0, when the motor does not receive the can command within a certain period of time, the motor enters the reset mode, and 20000 is 1s

3.3.7. Motor fault instructions

Function code 0x3022 indicates the fault code, where

bit14:motor blocking overload algorithm protection

bit7: Encoder uncalibrated: Motor uncalibrated encoder

bit3: Overvoltage fault: the motor voltage exceeds the protection voltage by 60V

bit2: Undervoltage fault: the motor voltage is lower than the protection voltage of 12V

bit1: Driver chip failure: Motor driver chip failure reported

bit0: Motor overtemperature fault: motor thermistor temperature exceeds 145

degrees

Function code 0x3024 is driver chip fault code 1. The specific faults are as follows

Table 11. Fault Status Register 1 Field Descriptions

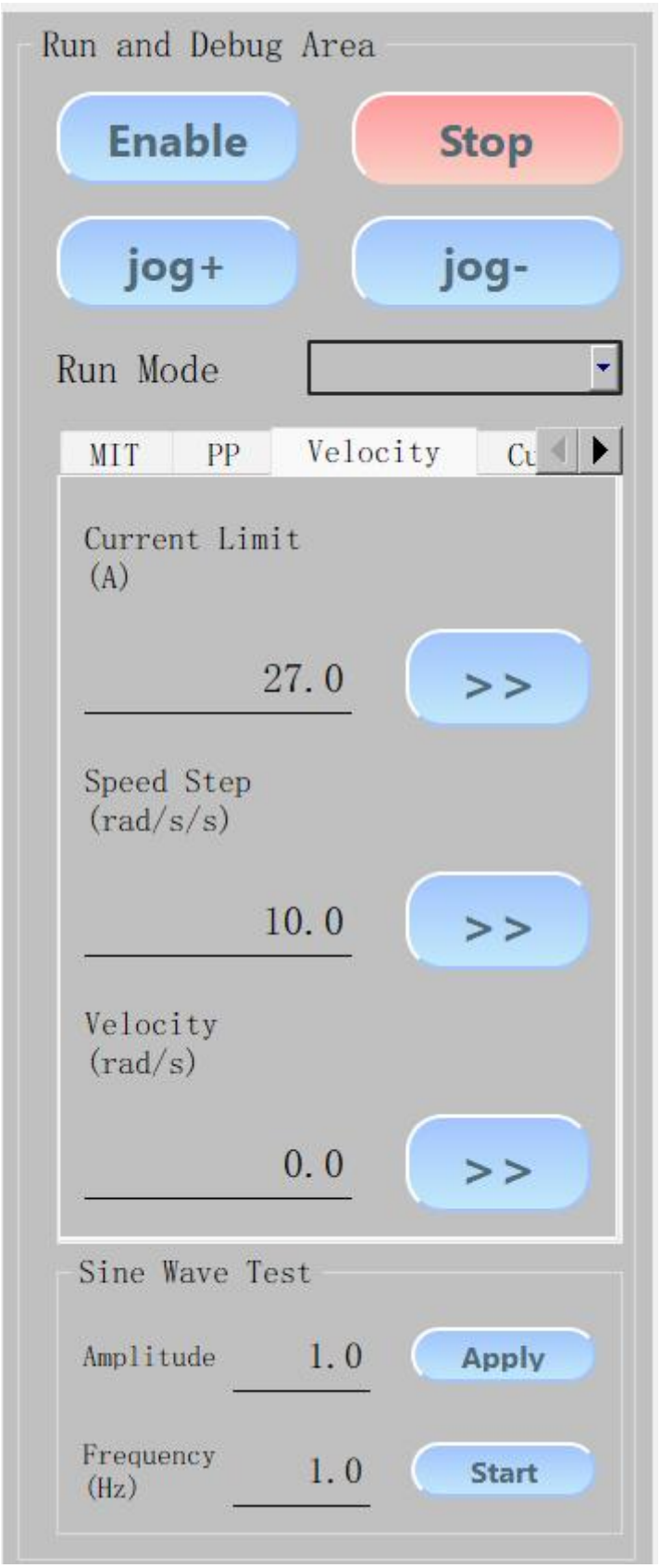
Bit	Field	Type	Default	Description
10	FAULT	R	0b	Logic OR of FAULT status registers. Mirrors nFAULT pin.
9	VDS_OCP	R	0b	Indicates VDS monitor overcurrent fault condition
8	GDF	R	0b	Indicates gate drive fault condition
7	UVLO	R	0b	Indicates undervoltage lockout fault condition
6	OTSD	R	0b	Indicates overtemperature shutdown
5	VDS_HA	R	0b	Indicates VDS overcurrent fault on the A high-side MOSFET
4	VDS_LA	R	0b	Indicates VDS overcurrent fault on the A low-side MOSFET
3	VDS_HB	R	0b	Indicates VDS overcurrent fault on the B high-side MOSFET
2	VDS_LB	R	0b	Indicates VDS overcurrent fault on the B low-side MOSFET
1	VDS_HC	R	0b	Indicates VDS overcurrent fault on the C high-side MOSFET
0	VDS_LC	R	0b	Indicates VDS overcurrent fault on the C low-side MOSFET

Function code 0x3025 is driver chip fault code 2. The specific faults are as follows

Table 12. Fault Status Register 2 Field Descriptions

Bit	Field	Type	Default	Description
10	SA_OC	R	0b	Indicates overcurrent on phase A sense amplifier (DRV8353xS)
9	SB_OC	R	0b	Indicates overcurrent on phase B sense amplifier (DRV8353xS)
8	SC_OC	R	0b	Indicates overcurrent on phase C sense amplifier (DRV8353xS)
7	OTW	R	0b	Indicates overtemperature warning
6	GDUV	R	0b	Indicates VCP charge pump and/or VGLS undervoltage fault condition
5	VGS_HA	R	0b	Indicates gate drive fault on the A high-side MOSFET
4	VGS_LA	R	0b	Indicates gate drive fault on the A low-side MOSFET
3	VGS_HB	R	0b	Indicates gate drive fault on the B high-side MOSFET
2	VGS_LB	R	0b	Indicates gate drive fault on the B low-side MOSFET
1	VGS_HC	R	0b	Indicates gate drive fault on the C high-side MOSFET
0	VGS_LC	R	0b	Indicates gate drive fault on the C low-side MOSFET

3.4. Control Demo



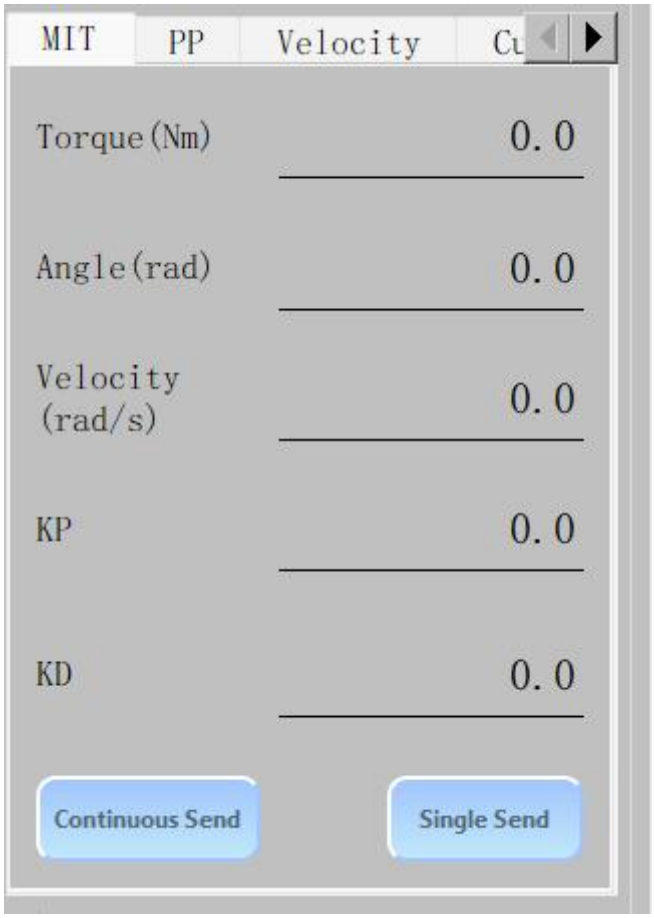
Jog Run

Click JOG +/- to run the motor forward and reverse at a speed of 1 rad/s.

Control Mode Switching

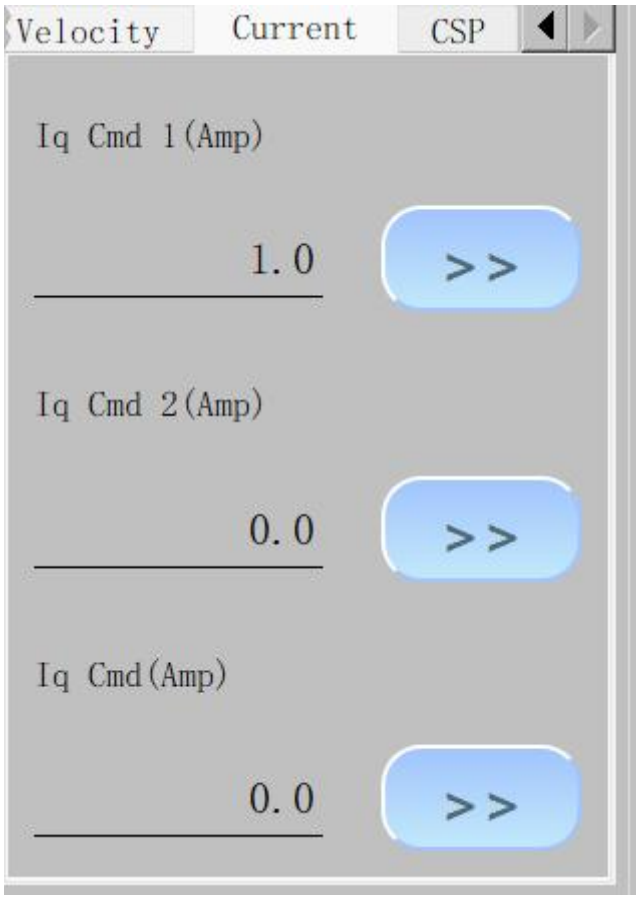
Select the desired control mode in the command box to the right of the run mode.

3.4.1. Operation and control mode

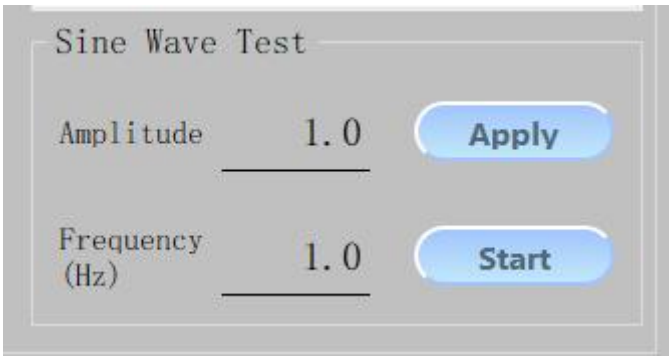


1. Switch the control mode to operation mode.
2. The motor starts running and enters motor_mode.
3. Set five parameter values and click Start or Send continuously. The motor will return feedback frames and run according to the target command.
4. Click Stop to stop the motor and terminate the continuous sending of commands.

3.4.2. Current mode



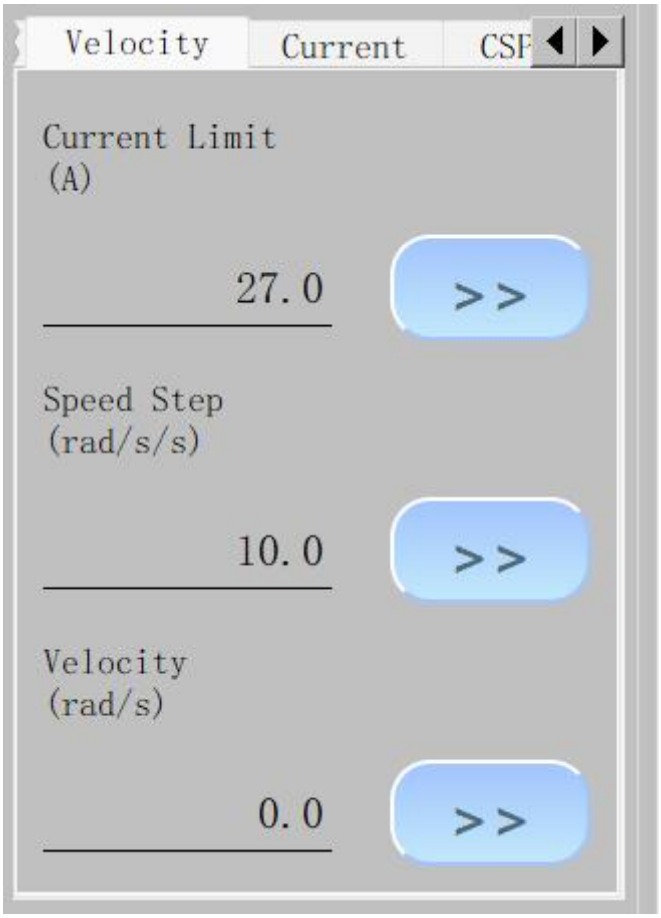
1. Switch the control mode to current mode.
2. The motor starts running and enters motor_mode.
3. Set the current command value for Iq Command 1 (A). Click the >> button on the right. The motor will follow the current command.
4. Click Stop to stop the motor.



Motor Current Sine Test

1. Switch the control mode to current mode.
2. The motor starts running and enters motor_mode.
3. Set the amplitude and frequency, click OK, and then click Start. The corresponding mode target command will be planned according to the sine law.
4. Click Stop to stop the motor.

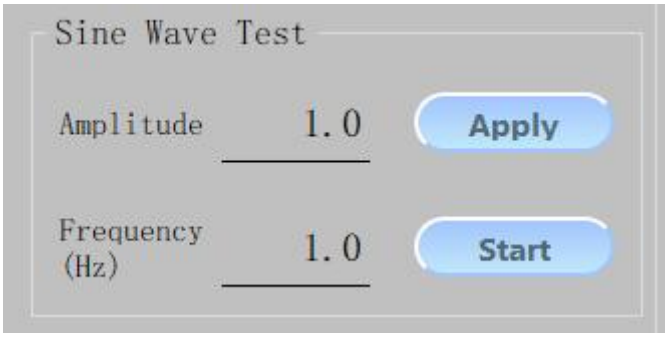
3.4.3. Speed Mode



1. Switch the control mode to speed mode.
2. The motor starts running and enters motor_mode.
3. First, set the current limit (maximum phase current) and speed step value (motor acceleration). If no settings are made, the motor will operate at the default values. Finally, set the speed command (target speed). The motor will

follow the command.

4. Click Stop to stop the motor.



Motor Speed Sine Test

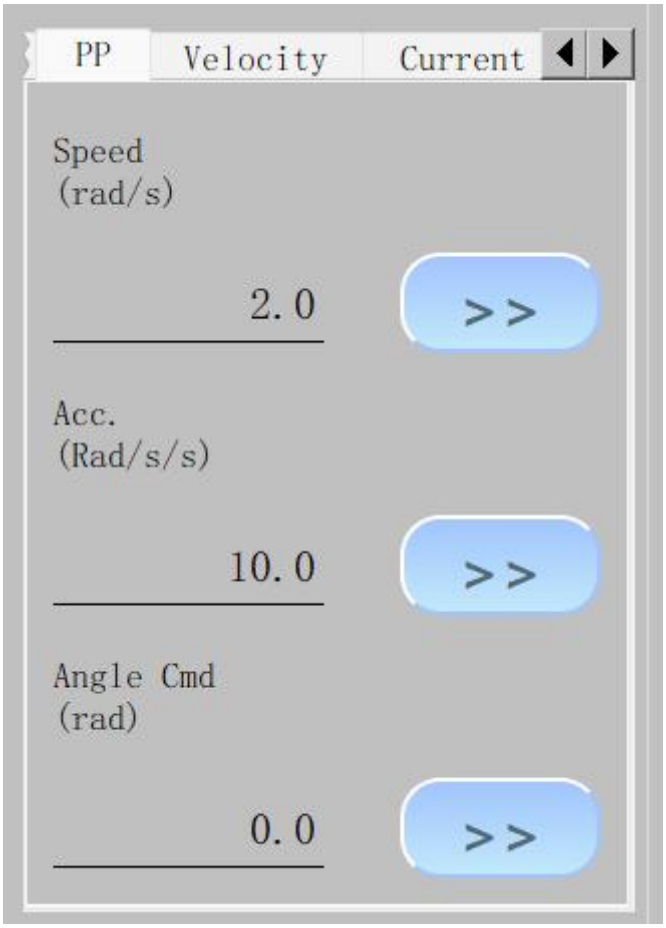
1. Switch the control mode to speed mode.

2. The motor starts running and enters motor_mode.

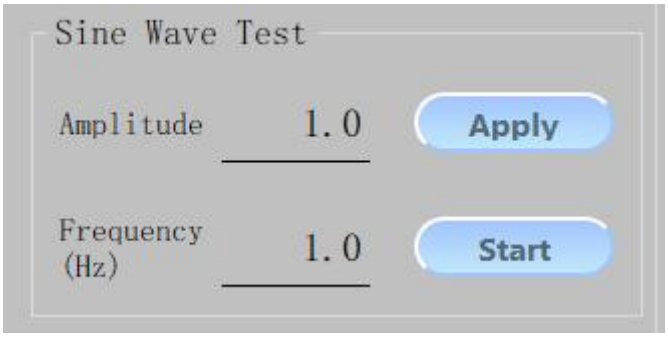
3. Set the amplitude and frequency, click OK, and then click Start. The corresponding mode target command will be planned according to the sine law.

4. Click Stop to stop the motor.

3.4.4. Position Mode (PP)



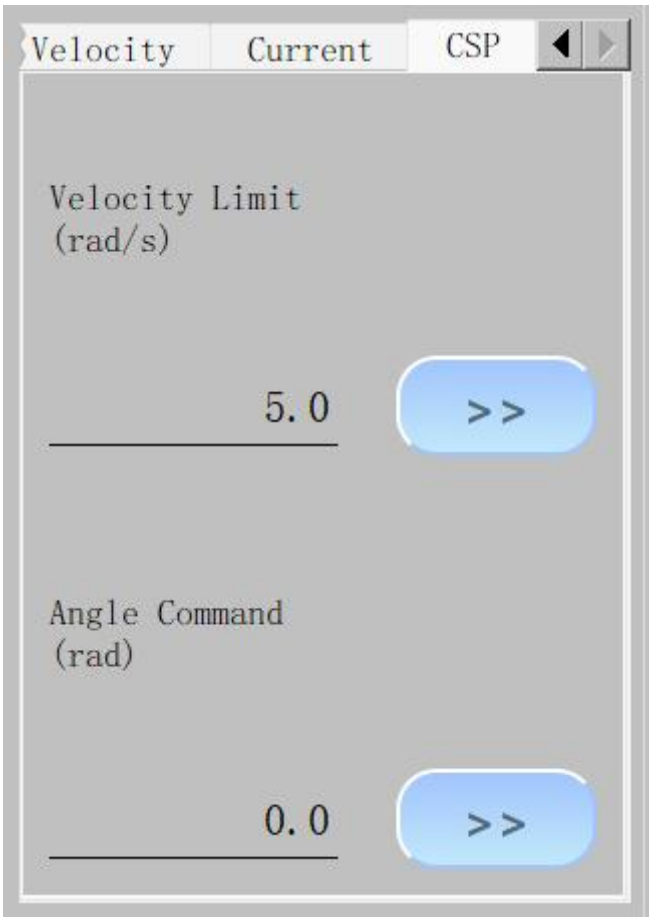
1. Switch the control mode to interpolation position mode.
2. Start the motor and enter motor_mode.
3. First, set the speed and acceleration. If not set, the motor will operate at the default values. Finally, set the position command (target position). The motor will follow the command.
4. Set the speed to 0 to stop the motor at the current position. To continue operation, re-issue the speed and position.
5. Click Stop to stop the motor.



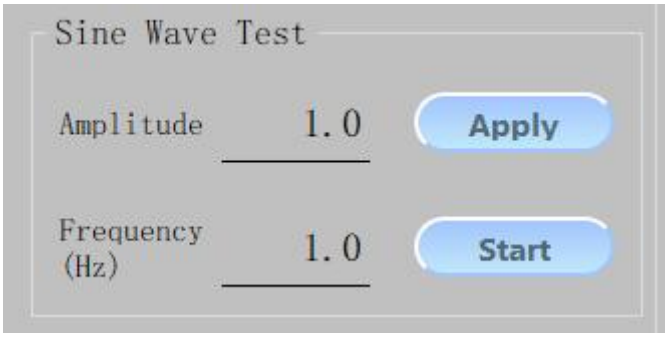
Motor Position Sine Test

1. Switch the control mode to interpolation position mode.
2. Start the motor and enter motor_mode.
3. Set the amplitude and frequency, click OK, and then click Start. The corresponding mode target command will be planned according to the sine law.
4. Click Stop to stop the motor.

3.4.5. Position Mode（CSP）



1. Switch the control mode to position mode.
2. The motor starts running and enters motor_mode.
3. Set the speed first. If no speed setting is made, the motor will run at the default value. Finally, set the position command (target position). The motor will follow the command.
4. Click Stop to stop the motor.



Motor position sinusoidal test

1. Switch the control mode to position mode.
2. The motor starts running and enters motor_mode.
3. Set the amplitude and frequency, click OK, and then click Start. The corresponding mode target command is then sinusoidally planned.
4. Click Stop to stop the motor.

4. Driver protocol and instructions

The motor communication is the CAN 2.0 communication interface, the baud rate is 1Mbps, and the extended frame format is adopted as follows:

data field	29-bit ID			8Byte data field
Size	Bit28~bit24	bit23~8	bit7~0	Byte0~Byte7
Description	Communication type	data area 2	Destination address	data area 1

The control modes supported by the motor include:

- Operation control mode: set 5 parameters of motor operation control;
- Current mode: the specified Iq current of the given motor;
- Velocity mode: the specified running speed of the given motor;
- Position mode: Given the specified position of the motor, the motor will run to the specified position;

4.1. Description of the communication protocol type

4.1.1. Communication type 0: Get device ID

Gets the device's ID and 64-bit MCU unique identifier

data field	29-bit ID			8Byte data field
Size	Bit28~bit24	bit23~8	bit7~0	Byte0~Byte7
Description	0x0	bit15~8: identifies host CAN_ID	target motor CAN_ID	0

Reply frame:

data field	29-bit ID			8Byte data field
Size	Bit28~bit24	bit23~8	bit7~0	Byte0~Byte7
Description	0x0	target motor CAN_ID	0XFE	64-bit MCU unique identifier

4.1.2. Communication Type 1: operation control mode motor control instruction

Data field	29 bit ID			8Byte data field
Size	Bit28~bit24	bit23~8	bit7~0	Byte0~Byte7
description	0x1	Byte2: Torque (0~65535) corresponds to (-60Nm~60Nm)	target motor CAN _ID	Byte0~1: target Angle [0~65535] corresponds to (-4π~4π) Byte2~3: Target angular velocity [0~65535] corresponds to (-20rad/s~20rad/s) Byte4~5: Kp [0~65535] corresponds to (0.0~5000.0) Byte6~7: Kd [0 to 65535] corresponds to the above data (0.0 to 100.0). After the conversion, the high byte is in front and the low byte is in

Response frame: Response motor feedback frame (see communication type 2)

Communication Type 2: motor feedback data

Data field	29 bit ID			8Byte data field
Size	Bit28~bit24	bit23~8	bit7~0	Byte0~Byte7
Description	0x2	Bit8~Bit15: CAN ID of the current motor bit21~16: fault information (0 none 1 has) bit21: uncalibrated bit20: Uncalibrated bit20: Gridlock overload fault bit19: magnetic coding fault bit18: overtemperature bit17: Three-phase overcurrent fault bit16: undervoltage fault bit22~23: Mode status 0: Reset mode [reset] 1: Cali mode [calibration] 2: Motor mode [Run]	host CAN _ID	Byte0~1: The current Angle [0~65535] Corresponding to (-4π~4π) Byte2~3: Current angular velocity [0~65535] corresponds to (-20rad/s~20rad/s) Byte4~5: Current torque [0~65535] corresponds to (-60Nm~60Nm) Byte6~7: Current temperature: Temp(Celsius) *10 If the value is higher than 10, the high byte is first and the low byte is last

4.1.3. Communication Type 3: Motor enabled to run

data field	29-bit ID			8Byte data field
Size	Bit28~bit24	bit23~8	bit7~0	Byte0~Byte7
Description	0x3	bit15~8: identifies the main CAN_ID	and target motor CAN_ID	

Response frame: Response motor feedback frame (see communication type 2)

4.1.4. Communication Type 4: Motor stops running

data field	29-bit ID			8Byte data field
Size	Bit28~bit24	bit23~8	bit7~0	Byte0~Byte7
description	0x4	bit15~8: used to identify the main CAN_ID	target motor CAN_ID	When the motor is running normally, 0 must be cleared in the data field. Byte[0]=1: The fault is cleared.

Response frame: Response motor feedback frame (see communication type 2)

Version number read frame

Data field	29 bit ID			8Byte data field
Size	Bit28~bit24	bit23~8	bit7~0	Byte0~Byte7
description	0x4	bit15~8: used to identify the main CAN_ID	target motor CAN_ID	Byte[0]=0x00 Byte[1]=0xC4

Response frame:

data field	29 位 ID			8Byte data field
Size	Bit28~bit24	bit23~8	bit7~0	Byte0~Byte7
description	0x2	Bit8~Bit15: CAN ID of the current motor bit21~16: fault information (0 none 1 has) bit21: uncalibrated bit20: Uncalibrated bit20: Gridlock overload fault bit19: magnetic coding fault bit18:overtemperature	target motor CAN_ID	Byte0=0x00 Byte1=0xC4 Byte2=0x56 Byte3~6:Motor version number Order from high to low

		bit17: Three-phase overcurrent fault bit16: undervoltage fault bit22~23: Mode status 0: Reset mode [reset] 1: Cali mode [calibration] 2: Motor mode [Run]		

4.1.5. Communication type 6: Set motor mechanical zero

data field	29-bit ID			8Byte data field
Size	Bit28~bit24	bit23~8	bit7~0	Byte0~Byte7
description	0x6	bit15~8: Identifies the main CAN_ID	and target motor CAN_ID	Byte[0]=1

Response frame: Response motor feedback frame (see communication type 2)

Communication type 7: Set motor CAN_ID

Change the current motor CAN_ID, effective immediately.

data field	29-bit ID			8Byte data field
Size	Bit28~bit24	bit23~8	bit7~0	Byte0~Byte7
description	0x7	bit15~8: used to identify main CAN_ID Bit16~23: preset CAN_ID	Target motor CAN_ID	

Answer frame: Answer motor broadcast frame (see communication type 0)

4.1.6. Communication type 17: Single parameter read

Data field	29 bit ID			8Byte data field
Size	Bit28~bit24	bit23~8	bit7~0	Byte0~Byte7
description	0x11	bit15~8: Used to identify the main CAN_ID	target motor CAN_ID	Byte0~1: index. For details, see the readability parameter table below Byte2~3:00 Byte4~7: In data above 00, the low byte is first and the high byte is second (

Reply frame:

Data field	29 bit ID			8Byte data field
Size	Bit28~bit24	bit23~8	bit7~0	Byte0~Byte7
description	0x11	bit15~8: indicates that the master CAN_ID Bit23~16:00 indicates that the master CAN_ID is successfully read. 01 indicates that the master can_ID	Byte0~1: Byte2~3:00 Byte4~7:Parameter data. 1 byte of data above Byte4 is preceded by low bytes and followed by high bytes at	

4.1.7. Communication type 18: Single parameter write (lost in power failure)

With type 22, the parameter starting with function code 0x20 of the parameter table in the upper computer module can be saved

Data field	29 bit ID			8Byte data field
Size	Bit28~bit24	bit23~8	bit7~0	Byte0~Byte7
description	0x12	bit15~8: Used to identify the main CAN_ID	target motor CAN_ID	Byte0~1: index. For details, see the readability parameter table below Byte2~3: 00 Byte4~7: Parameter data In the preceding data, the low byte is in the front and the high byte is in the rear

Response frame: Response motor feedback frame (see communication type 2)

4.1.8. Communication type 21: Fault feedback frame

data field	29-bit ID			8Byte data field
Size	Bit28~bit24	bit23~8	bit7~0	Byte0~Byte7
description	0x15	bit15~8: motor	identifies the main	Byte0~3: fault value (non-0: faulty; 0: faulty). Normal)

		CAN_ID	CAN_ID	Bit 16: A-phase current sampling overcurrent Bit 14: Motor stall overload algorithm protection Bit 9: Position initialization fault Bit 8: Hardware identification fault Bit 7: Encoder uncalibrated: Motor encoder not calibrated Bit 5: C-phase current sampling overcurrent Bit 4: B-phase current sampling overcurrent bit3: overvoltage fault bit2: undervoltage fault bit1: driver chip fault bit0: motor overtemperature fault, Default 145 ° C Byte4~7: warning Value bit0: motor overtemperature warning, the default is 135 ° c

4.1.9. Communication type 22: Motor data save frame

data field	29-bit ID			8Byte data field
Size	Bit28~bit24	bit23~8	bit7~0	Byte0~Byte7
Description	0x16	bit15~8: identifies the main CAN_ID	and target motor CAN_ID	01 02 03 04 05 06 07 08

Response frame: Response motor feedback frame (see communication type 2)

4.1.10. Communication type 23: Motor baud rate modification frame (re-power-on effect)

Data field	29 bit ID			8Byte data field
Size	Bit28~bit24	bit23~8	bit7~0	Byte0~Byte7
description	0x17	bit15~8: used to identify the main CAN_ID	target motor CAN_ID	01 02 03 04 05 06 F_CMD Among them, the F_CMD byte is the motor baud rate Among them, 01 is 1M 02 is 500K 03 is 250K 04 is 125K

Response frame: Response motor feedback frame (see communication type 0)

4.1.11. Communication type 24: The motor actively reports frames

data field	29-bit ID			8Byte data field
Size	Bit28~bit24	bit23~8	bit7~0	Byte0~Byte7
description	0x18	bit15~8: identifies the main CAN_ID	target motor CAN_ID	01 02 03 04 05 06 F_CMD Among them, the F_CMD byte is the motor reporting switch 00 is to disable active reporting (default) 01 To enable active reporting, the default reporting interval is 10ms

Response frame:

data field	29 位 ID			8Byte data field
Size	Bit28~bit24	bit23~8	bit7~0	Byte0~Byte7
description	0x18	Bit8~Bit15: CAN ID of the current motor bit21~16: fault information (0 none 1 has) bit21: uncalibrated bit20: Uncalibrated bit20: Gridlock overload fault bit19: magnetic coding fault bit18:overtemperature bit17: Three-phase overcurrent fault bit16: undervoltage fault bit22~23: Mode status 0: Reset mode [reset] 1: Cali mode [calibration] 2: Motor mode [Run]	target motor CAN_ID	Byte0~1: The current Angle [0~65535] Corresponding to (-4π ~4π) Byte2~3: Current angular velocity [0~65535] corresponds to (-20rad/s~20rad/s) Byte4~5: Current torque [0~65535] corresponds to (-60Nm~60Nm) Byte6~7: Current temperature: Temp(Celsius) *10 If the value is higher than 10, the high byte is first and the low byte is last

4.1.12. Communication type 25: Motor protocol modification frame (re-power-on effect)

Data field	29 bit ID			8Byte data field
Size	Bit28~bit24	bit23~8	bit7~0	Byte0~Byte7
description	0x19	bit15~8: used to identify the main CAN_ID	target motor CAN_ID	01 02 03 04 05 06 F_CMD Among them, the F_CMD byte is the motor protocol type Among them, 0 is a private protocol (default) 1 is the Canopen protocol 2 is the MIT protocol

Response frame: Response motor feedback frame (see communication type 0)

4.1.13. Read and write a single parameter list

index		Description	Type	Number of bytes		R/W Read and write permission
0X7005	run_mode	0: operation mode 1: position mode (PP) 2: Velocity mode 3: Operation mode Current mode 5: Position mode (CSP)	uint8	1		W/R
0X7006	iq_ref	Current mode Iq command	float	4	-43 to 43A	W/R
0X700A	spd_ref	Rotational Velocity mode Rotational speed command	float	4	-20to 20rad/s	W/R
0X700B	limit_torque	torque limit	float	4	0 to 60Nm	W/R
0X7010	cur_kp	Kp	float	4	The default value is 0.17	W/R
0X7011	cur_ki	Ki	float	4	The default value is 0.012	W/R
0X7014	cur_filt_gain	filt_gain	float	4	0 to 1.0, The default value is 0.1	W/R
0X7016	loc_ref	Position Mode Angle instruction	float	4	rad	W/R
0X7017	limit_spd	Location mode (CSP) speed limit	float	4	0 to 20rad/s	W/R
0X7018	limit_cur	Velocity position mode Current limitation	float	4	0 to 43A	W/R
0x7019	mechPos	Mechanical Angle of the loading coil	float	4	rad	R
0x701A	iqf	iq Filter	float	4	-43 to 43A	R
0x701B	mechVel	Speed of the load	float	4	-20 to	R

index		Description	Type	Number of bytes		R/W Read and write permission
					20rad/s	
0x701C	VBUS	Bus voltage	float	4	V	R
0x701E	loc_kp kp	at	float	4	The default value is 60	W/R
0x701F	spd_kp	Indicates the speed kp	float	4	The default value is 6	W/R
0x7020	spd_ki	ki	float	4	The default value is 0.02	W/R
0x7021	spd_filt_gain	Speed filter value	float	4	The default value is 0.1	W/R
0x7022	acc_rad	velocity mode acceleration	float	4	The default value is 20rad/s^2	W/R
0x7024	vel_max	Location mode (PP) speed	float	4	The default value is 10rad/s	W/R
0x7025	acc_set	Location mode (PP) acceleration	float	4	The default value is 10rad/s^2	W/R
0x7026	EPScan_time	Indicates the report time. 1 indicates 10ms. Plus 1 increments by 5ms	uint16	2	The default value is 1	W/R
0x7028	canTimeout	can The timeout threshold, 20000 is 1s	uint32	4	The default value is 0	W/R
0x7029	zero_sta	Indicates the zero flag bit, 0 means 0-2π and 1 means -π-π	uint8	1	The default is 0	W/R
0x702A	damper	Damping switch	uint8	1	The default value is 0	W/R
0x702B	add_offset	Zero offset	float	4	The default is 0	W/R
0x702C	alveolous_open	Cogging Compensation Switch	uint8	1	The default is 0	W/R

index		Description	Type	Number of bytes		R/W Read and write permission
0x702D	iq_test	Motor Initialization Calibration Switch	uint8	1	The default is 0	W/R
0x702E	dcc_set	Deceleration in Motor PP Mode (Normally Disabled)	float	4	The default is 10rad/s^2	W/R

4.1.14. Read example:

Take reading loc_kp as an example:

Read instruction is

Size	Bit28~bit24	bit23~8	bit7~0	Byte0~Byte7
	0x11	0x00FD	0x7F	1E 70 00 00 00 00 00 00
Description	Type 17	Host id 0xFD	Target motor CAN_ID 7F	Byte0~1: index, corresponding to loc_kp

The feedback instruction is

Size	Bit28~bit24	bit23~8	bit7~0	Byte0~Byte7
	0x11	0x007F	0xFD	1E 70 00 00 00 00 F0 41
Description	Type 17	bit15~8: Target motor CAN_ID 7F	Host id 0xFD	Byte0~1: index, corresponding to loc_kp Byte4~7:loc_kp value 30, high right byte, (32-bit single precision) hexadecimal IEEE-754 standard floating point number

4.2. Motor Function Description

(If the following features are unavailable, please upgrade to the latest version via the official Git repository.)

4.2.1. Active Reporting

- **Disabled by default**. Enable via **Type 24**.

- Report type: **Type 2** (default interval: **10ms**). Adjust interval by modifying `EPScan_time` via **Type 18**.

4.2.2. Zero-Point Flag (`zero_sta`)

- Modify via:
  - **Host computer**
  - **Type 18** (*requires saving via **Type 22** for communication*)
- **Default flag**: `0` → Power-on position range: **0–2π**.
- **If set to `1`**: Power-on position range: **π–π**.

4.2.3. Type 2 Update

- Updated to **periodic looping within -4π–4π** (enables cycle counting).
- **Note**: Position interface parameters must be adjusted:
  - `P_MIN`: **12.57f**
  - `P_MAX`: **12.57f**

4.2.4. Protocol Switching *(Requires CAN adapter)*

- Methods:
  - Modify `protocol_1` via host computer.
  - Send **Type 25** command.
- **Reboot required** after switching.
- **Post-switch CAN commands**:
  - **CANopen**: Send extended frame (protocol switch frame).
  - **MIT Protocol**: Send standard frame (**Command 8**).

4.2.5.Motor Post-Power-Off Anti-Backdrive Protection

- **The motor’s post-power-off anti-backdrive protection can be enabled or disabled.If the motor is rotated rapidly while unpowered, a certain damping force will be felt. This protection prevents surges caused by excessively high external rotation speeds.The protection can be disabled by setting damper to 1.**

4.2.6. Zero Calibration Rules

- **Zero calibration is supported in CSP and Motion Control modes, but blocked in PP mode.**
- **In old firmware versions, zero calibration would cause a large deviation between the motor's target and actual position, triggering an immediate movement of the motor to the target position.**
- **In new firmware versions, when zero calibration is performed in CSP or Motion Control mode, the motor's target position is instantly updated to 0, so the motor remains stationary. Zero calibration is unavailable in PP mode.**

4.2.7.Position Offset

- **A position offset can be added. If an offset of 1 is set, the new zero point will be the original zero point plus 1.**
- **Some customers cannot perform zero calibration at the physical zero position due to mechanical limits.For example, if the zero point is set at a position of 1 radian, and add_offset is then set to 1, after a power cycle, the motor’s position of 1 radian will correspond to the original 1 radian position, and the motor’s zero point will be 1 radian counterclockwise from the original 1 radian position.**

4.2.8. The CANopen ID of the motor is consistent with the CAN ID.

- **The CANopen ID is kept the same as the CAN ID under the private protocol.**

4.2.9. Motor Initialization Calibration

- When the motor sets iq_test to 1, the initialization time is extended, resulting in a more accurate motor initial reference.

4.2.10. Cogging Calibration Function

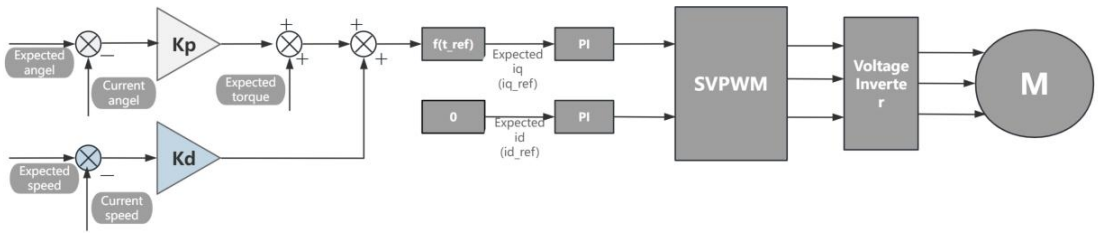
- Before calibration, set iq_test to 1 and restart the motor; the motor must be under no-load conditions. Then perform cogging calibration via the host computer. After successful calibration, max_alve will be non-zero. Next, set alveolous_open to 1 to activate the function in Motion Control mode.

4.2.11. PP Mode Deceleration

By setting acc_stutus to 1, the acceleration and deceleration for PP mode can be configured separately. The deceleration value is stored at 0x702E. When acc_stutus is set to 0, the acceleration parameter controls both acceleration and deceleration.

4.3. Control mode instructions

4.3.1. Operation control mode

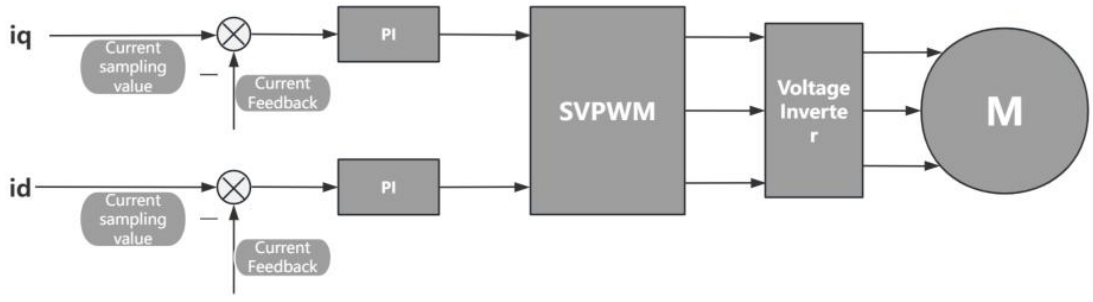


The motor is in operation control mode by default after power-on.

Send motor Enable Run frame (communication type 3) --> Send operation mode motor control command (communication type 1) --> Receive motor feedback frame (communication type 2)

Operation control mode description: The control logic of the operation and control mode is t_ref=Kd * (v_vset-v_actual)+Kp * (p_set-p_actual)+t_ff. Tref is converted to the expected iq current through an internal formula and output through the current loop Simple control demonstration: Set t_ff to 0, v_vset to 1, Kd to 1, p_set to 0, Kp to 0. If there is no external load on the motor, it will run at a speed of 1rad/s. If there is an external load, kd needs to be increased to resist the external load Set t_ff to 0, v_vset to 0, Kd to 1, p_set to 0, Kp to 0, the motor is in damping mode. When the motor is externally rotated, a damping is applied, which increases with the increase of kd. It should be noted that the motor generates electricity under this condition and requires power supply to prevent overvoltage Set t_ff to 0, v_vset to 0, Kd to 1, p_set to 5, Kp to 1. If there is no external load on the motor, it will run to the target position of 5. Increasing kp will increase the force required to maintain the target position, and kd is damping. Without kd, the motor will sway to the target position

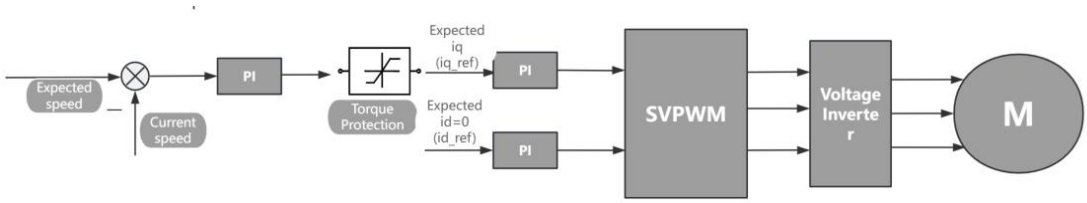
4.3.2. Current mode



Send motor mode parameter write command (communication type 18) Set the runmode parameter to 3 --> Send motor Enable run frame (communication type 3) --> Send motor mode parameter write command (communication

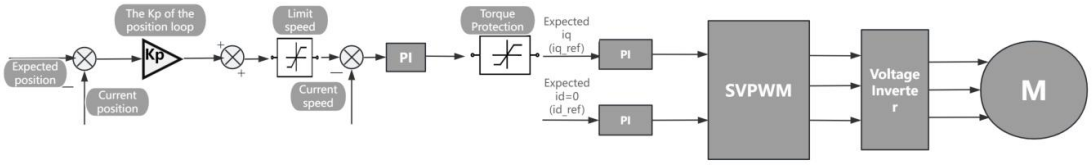
type 18) set the iq_ref parameter to the default current instruction

4.3.3. Velocity mode



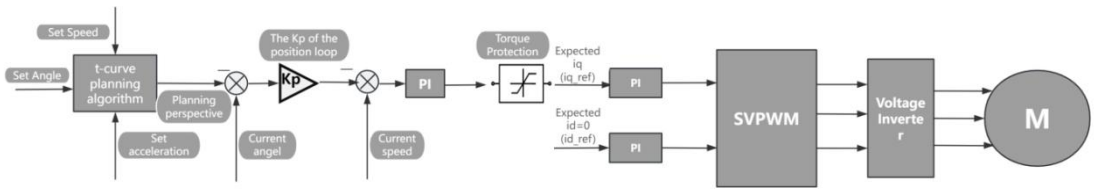
Send motor mode parameter write command (communication type 18) Set the runmode parameter to 2 --> Send motor Enable run frame (communication type 3) --> Send motor mode parameter write command (communication type 18) set limit_cur parameter as default maximum current instruction --> Send motor mode parameter write command (communication type 18) Set acc_rad parameter as default acceleration instruction --> Send motor mode parameter write command (communication type 18) Set spd_ref parameter as default speed instruction

4.3.4. Location Mode (CSP)



Send motor mode parameter write command (communication type 18) Set the runmode parameter to 5 --> Send motor Enable run frame (communication type 3) --> Send motor mode parameter write command (communication type 18) set limit_spd parameter as default maximum speed instruction --> Send motor mode parameter write command (communication type 18) Sets loc_ref parameter as default position instruction

4.3.5. Location Mode (PP)



Send motor mode parameter write command (communication type 18) Set the runmode parameter to 1 --> Send motor Enable run frame (communication type 3) --> Send motor mode parameter write command (communication type 18) set The vel_max parameter is the default maximum speed instruction --> Send motor mode parameter write command (communication type 18) Set the acc_set parameter to the default acceleration instruction --> Send motor mode parameter write command (communication type 18) Set the loc_ref parameter to the default position instruction

Note: This mode does not support changing the speed and acceleration during operation. If you want to make an emergency stop, you can change vel_max to 0 during the process, and it will stop at the current speed and acceleration plan

4.3.6. Stop running

Sending motor stop frame (communication type 4)

4.4. Program sample

Examples of various mode control motors are provided below (take gd32f303 as an example)

The following are library, function, and macro definitions for the various instances

```
Plain Text
```

```
#define P_MIN -12.57f

#define P_MAX 12.57f

#define V_MIN -20.0f

#define V_MAX 20.0f

#define KP_MIN 0.0f

#define KP_MAX 5000.0f

#define KD_MIN 0.0f

#define KD_MAX 100.0f

#define T_MIN -60.0f

#define T_MAX 60.0f

struct exCanIdInfo{

uint32_t id:8;

uint32_t data:16;

uint32_t mode:5;

uint32_t res:3;
```

```
};

can_receive_message_struct rxMsg;

can_trasnmit_message_struct txMsg={

.tx_sfid = 0,

.tx_efid = 0xff,

.tx_ft = CAN_FT_DATA,

.tx_ff = CAN_FF_EXTENDED,

.tx_dlen = 8,

};

#define txCanIdEx (*((struct exCanIdInfo*)&(txMsg.tx_efid)))

#define rxCanIdEx (*((struct exCanIdInfo*)&(rxMsg.rx_efid))) //将扩展帧 id 解析为自定义数据结构

int float_to_uint(float x, float x_min, float x_max, int bits){

float span = x_max - x_min;

float offset = x_min;

if(x > x_max) x=x_max;
```

```
else if(x < x_min) x= x_min;

return (int) ((x-offset)*((float)((1<<bits)-1))/span);

}

#define can_txd() can_message_transmit(CAN0, &txMsg)

#define can_rxd() can_message_receive(CAN0, CAN_FIFO1, &rxMsg)
```

The following lists the common types of communication sent:

4.4.1. Motor Enabled Run frame (communication type 3)

```
Plain Text
void motor_enable(uint8_t id, uint16_t master_id)
{
txCanIdEx.mode = 3;
txCanIdEx.id = id;
txCanIdEx.res = 0;
txCanIdEx.data = master_id;
txMsg.tx_dlen = 8;
txCanIdEx.data = 0;
can_txd();
}
```

4.4.2. Operation control mode Motor control instruction (communication type 1)

Plain Text

```
void motor_controlmode(uint8_t id, float torque, float MechPosition, float speed, float kp, float kd)

{

txCanIdEx.mode = 1;

txCanIdEx.id = id;

txCanIdEx.res = 0;

txCanIdEx.data = float_to_uint(torque,T_MIN,T_MAX,16);

txMsg.tx_dlen = 8;

txMsg.tx_data[0]=float_to_uint(MechPosition,P_MIN,P_MAX,16)>>8;

txMsg.tx_data[1]=float_to_uint(MechPosition,P_MIN,P_MAX,16);

txMsg.tx_data[2]=float_to_uint(speed,V_MIN,V_MAX,16)>>8;

txMsg.tx_data[3]=float_to_uint(speed,V_MIN,V_MAX,16);

txMsg.tx_data[4]=float_to_uint(kp,KP_MIN,KP_MAX,16)>>8;

txMsg.tx_data[5]=float_to_uint(kp,KP_MIN,KP_MAX,16);
```

```
txMsg.tx_data[6]=float_to_uint(kd,KD_MIN,KD_MAX,16)>>8;

txMsg.tx_data[7]=float_to_uint(kd,KD_MIN,KD_MAX,16);

can_txd();

}
```

4.4.3. Motor mode parameter write command (communication type 18, running mode switch)

Plain Text

```
void motor_reset(uint8_t id, uint16_t master_id)

{

txCanIdEx.mode = 4;

txCanIdEx.id = id;

txCanIdEx.res = 0;

txCanIdEx.data = master_id;

txMsg.tx_dlen = 8;

for(uint8_t i=0;i<8;i++)

{
```

```
txMsg.tx_data[i]=0;

}

can_txd();

}
```

4.4.4. Motor mode parameter write command (communication type 18, running mode switch)

Plain Text

```
uint8_t runmode;

uint16_t index;

void motor_modechange(uint8_t id, uint16_t master_id)

{

txCanIdEx.mode = 0x12;

txCanIdEx.id = id;

txCanIdEx.res = 0;

txCanIdEx.data = master_id;

txMsg.tx_dlen = 8;
```

```
for(uint8_t i=0;i<8;i++)

{

txMsg.tx_data[i]=0;

}

memcpy(&txMsg.tx_data[0],&index,2);

memcpy(&txMsg.tx_data[4],&runmode, 1);

can_txd();

}
```

4.4.5. Motor mode parameter write command (communication type 18, control parameter write)

Plain Text

```
uint16_t index;

float ref;

void motor_write(uint8_t id, uint16_t master_id)

{
```

```
txCanIdEx.mode = 0x12;

txCanIdEx.id = id;

txCanIdEx.res = 0;

txCanIdEx.data = master_id;

txMsg.tx_dlen = 8;

for(uint8_t i=0;i<8;i++)

{

txMsg.tx_data[i]=0;

}

memcpy(&txMsg.tx_data[0],&index,2);

memcpy(&txMsg.tx_data[4],&ref,4);

can_txd();

}
```

5. Explanation of Canopen Communication Protocol Types

5.1. Introduction to CANopen Communication

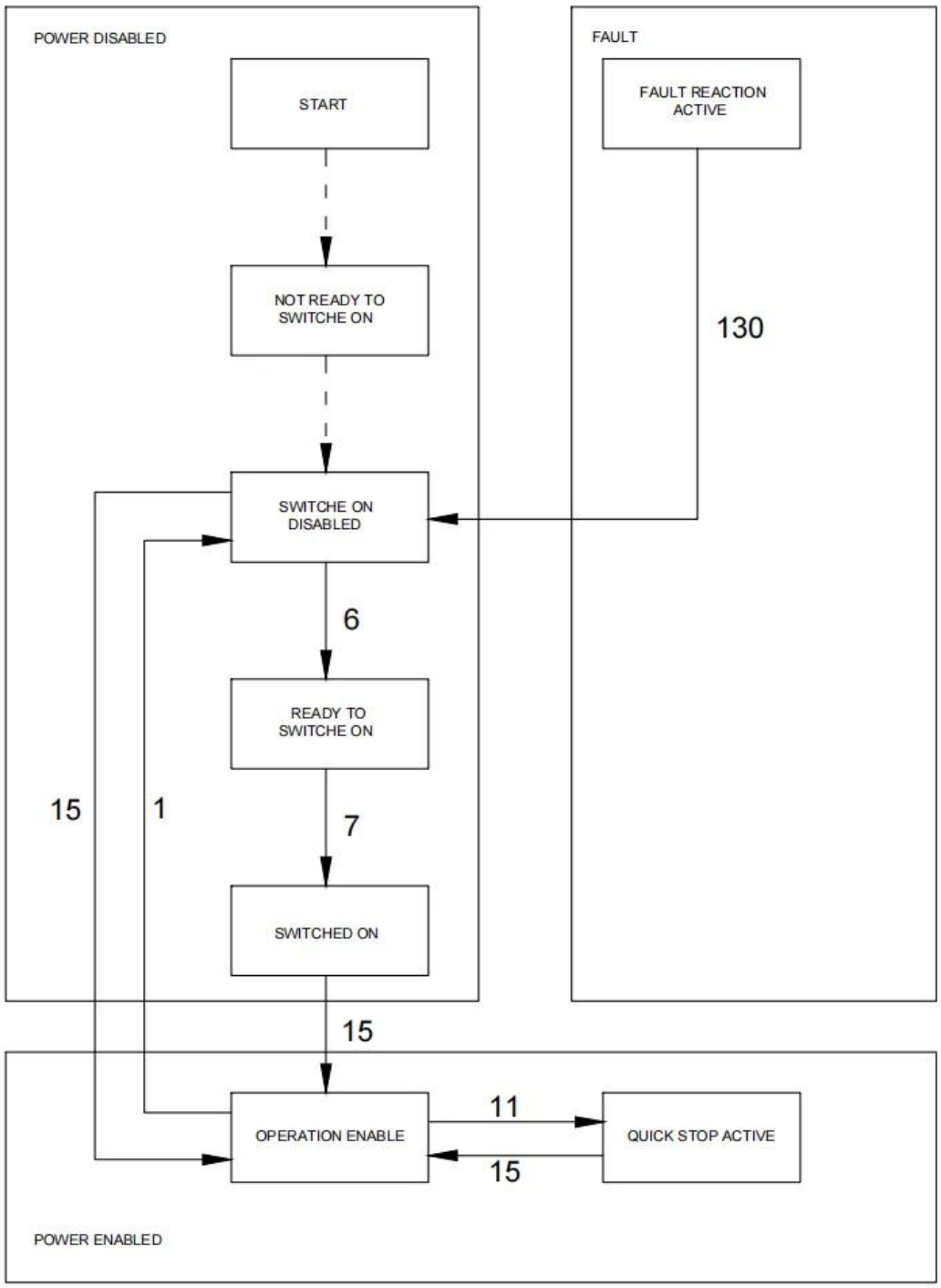
CANopen is a "higher-level protocol" based on the CAN bus. This means that the CAN bus (ISO 11898) acts like a container truck, serving as the "transportation vehicle" for CANopen information. CAN simply implements the transmission of frames with an 11-bit CAN ID, a Remote Transmission (RTR) bit, and 64 data bits (related to higher-level protocols). The CAN bus plays the same role in CANopen as it does in the J1939 protocol. CANopen implements Layer 7 of the OSI model and is compatible with other data link layer protocols besides CAN.Please download the necessary EDS files for CANopen from the official website download center at www.robstride.com.

5.2. Canopen Protocol Message Classification

Classification	Function
NMT Network Management Messages	Manage the network and switch node states. NMT network management messages are typically sent by the master station.
SDO Service Data Object Message	Used for setting device parameters or transmitting critical data. Typically, the master station initiates an SDO message, and the slave station responds. However, a slave station can also initiate an SDO message, with the master station responding, for example, in the transmission of critical data.
PDO (Process Data Object) messages	Transmit process data from various devices, such as temperature and speed. Both the master and slave stations send these messages.
EMCY Emergency Message	Transmits fault information about the transmission equipment. Both the master and slave stations send this message.
SYNC Synchronization Message	Synchronization data, used to synchronize TPDO data on the slave station. Generally sent by the master station.

Classification	Function
NODE GUARDING Message	The master station requests the slave station's status; the master station queries, and the slave station responds.
Heartbeat Message	A device actively sends a heartbeat to indicate that it is online. Both the master and slave devices can send heartbeats.

5.3. State Machine Description



Motor Enable:

When initially powered on, the motor defaults to the **SWITCH_ON_DISABLED** state. To transition to **OPERATION_ENABLE**, modify the **Controlword (6040H)** to **6, 7, or 15** (step-by-step transition), or directly set it to **15** for immediate

enablement.

Stopping the Motor:

If the motor is in **OPERATION_ENABLE** state and needs to stop normally, modify the **Controlword (6040H)** to **1**. The motor will return to the disabled state (**SWITCH_ON_DISABLED**).

Emergency Stop (Use with Caution—Risk of Voltage Surge):

During operation, an emergency stop can be triggered by setting the **Controlword (6040H)** to **11**.

Fault Clearance:

If the motor enters a **FAULT** state due to protection mechanisms, modifying the **Controlword (6040H)** can clear standard errors.

Motor Watchdog

Set the CAN communication watchdog via index 1 of object 0x6099. After configuration, a value of 20000 represents 1 second. If no communication frame is received within the set time, the motor will be disabled. The default value is 0 (disabled), and a non-zero value enables the watchdog.

Important Note:

Mode changes for this motor must be performed in the **disabled state (SWITCH_ON_DISABLED)**. Ensure the desired mode is configured **before** enabling **OPERATION_ENABLE** to avoid unexpected behavior.

5.4. Status Feedback Parameters

Index	Name	Attribute	Type	Unit
603F	**Error_code**	Read-only	UINTEGER16	/
6041	**Statusword**	Read-only	UINTEGER16	/
6061	**Modes_of_operation_display**	Read-only	INTEGER8	/
6062	**Position_demand_value**	Read-only	INTEGER32	Pulses (1 rev = 16,384 pulses)

Index	Name	Attribute	Type	Unit
6064	**Position_actual_value**	Read-only	INTEGER32	Pulses (1 rev = 16,384 pulses)
606B	**Velocity_demand_value**	Read-only	INTEGER32	0.1 rpm
606C	**Velocity_actual_value**	Read-only	INTEGER32	0.1 rpm
6077	**Torque_actual_value**	Read-only	INTEGER16	0.1% load ratio (1000 = 20 N·m)
6078	**Current_actual_value**	Read-only	INTEGER16	mA
6079	**DC_link_circuit_voltage**	Read-only	INTEGER32	mV

5.5. Homing Mode (Zero Position Setting)

Index	Name	Attribute	Type	Unit
6040	**Controlword**	Read-write	UINTEGER16	/
6060	**Modes of operation**	Read-write	INTEGER8	/

Homing method:

- Set **Modes of operation** to **6** while the motor is in the **disabled state (SWITCH_ON_DISABLED)**. The motor will then define the current position as the zero point.
- To **hold the zero position**, modify the **Controlword** to **15**, and the motor will maintain its position at the home location.

5.6. Position Mode (PP - Profile Position)

Index	Name	Attribute	Type	Unit
6040	**Controlword**	Read-write	UINTEGER16	/
6060	**Modes of operation**	Read-write	INTEGER8	/
6067	**Position_window**	Read-write	UINTEGER32	Pulses (1 rev = 16,384 pulses)
6068	**Position_window_time**	Read-write	UINTEGER16	ms
6071	**Target_torque**	Read-	INTEGER16	0.1% load ratio (1000 = 20

Index	Name	Attribute	Type	Unit
		write		N·m)
607A	**Target_position**	Read-write	INTEGER32	Pulses (1 rev = 16,384 pulses)
6081	**Profile_velocity**	Read-write	UINTEGER32	0.1 rpm
6083	**Profile_acceleration**	Read-write	UINTEGER32	0.1 rpm/s

Steps to Configure Position Mode (PP):

1. While the motor is in the **disabled state (SWITCH_ON_DISABLED)**, set **Modes of operation** to **1**.
    - **Mandatory parameters**:
        - **Target_torque** (absolute max torque in position mode)
        - **Profile_velocity** (absolute speed in position mode)
        - **Profile_acceleration** (absolute acceleration in position mode)
    - **Optional parameters**:
        - **Position_window** (if not set, window check is disabled)
        - **Position_window_time** (if not set, window check is disabled)
2. Set **Controlword (6040)** to **15** to enable operation.
3. **Set Target_position** (absolute position) to move the motor to the desired position.

5.7. Position Mode (CSP - Cyclic Synchronous Position)

Index	Name	Attribute	Type	Unit
6040	**Controlword**	Read-write	UINTEGER16	/
6060	**Modes of operation**	Read-	INTEGER8	/

Index	Name	Attribute	Type	Unit
		write		
6067	**Position_window**	Read-write	UINTEGER32	Pulses (1 rev = 16,384 pulses)
6068	**Position_window_time**	Read-write	UINTEGER16	ms
6071	**Target_torque**	Read-write	INTEGER16	0.1% load ratio (1000 = 20 N·m)
607A	**Target_position**	Read-write	INTEGER32	Pulses (1 rev = 16,384 pulses)
6081	**Profile_velocity**	Read-write	UINTEGER32	0.1 rpm

Steps to Configure Position Mode (CSP):

1. While the motor is in the **disabled state (SWITCH_ON_DISABLED)**, set **Modes of operation** to **5**.
    - **Mandatory parameters**:
        - **Target_torque** (absolute max torque in position mode)
        - **Profile_velocity** (absolute speed in position mode)
    - **Optional parameters**:
        - **Position_window** (0 = disabled)
        - **Position_window_time** (0 = disabled)
2. Set **Controlword (6040)** to **15** to enable operation.
3. **Set Target_position** (absolute position) to move the motor to the desired position.

5.8. Velocity Mode

Index	Name	Attribute	Type	Unit
6040	**Controlword**	Read-write	UINTEGER16	/
6060	**Modes of operation**	Read-write	INTEGER8	/
6071	**Target_torque**	Read-write	INTEGER16	0.1% load ratio (1000 = 20

Index	Name	Attribute	Type	Unit
				N·m)
60FF	**Target_velocity**	Read-write	INTEGER32	0.1 rpm

Steps to Configure Velocity Mode:

1. While the motor is in the **disabled state (SWITCH_ON_DISABLED)**, set **Modes of operation** to **3**.
    - **Mandatory parameter**:
        - **Target_torque** (absolute max torque in velocity mode)
2. Set **Controlword (6040)** to **15** to enable operation.
3. **Set Target_velocity** to reach the desired speed.

5.9. Torque Mode

Index	Name	Attribute	Type	Unit
6040	**Controlword**	Read-write	UINTEGER16	/
6060	**Modes of operation**	Read-write	INTEGER8	/
6071	**Target_torque**	Read-write	INTEGER16	0.1% load ratio (1000 = 20N·m)

Steps to Configure Torque Mode:

1. While the motor is in the **disabled state (SWITCH_ON_DISABLED)**, set **Modes of operation** to **4**.
2. Set **Controlword (6040)** to **15** to enable operation.
3. **Set Target_torque** to output the desired torque.

5.10. Protocol Switching (Extended Frame): Switch Motor Protocol (Takes Effect After Power Cycle)

Data Field	29-bit ID	8-Byte Data Area

Size	Bit 28~0	Byte 0~6
Description	**0xFFF**	01 02 03 04 05 06 F_CMD

- **F_CMD** (Byte 6) defines the motor protocol:
  - **0**: Private protocol (default)
  - **1**: CANopen protocol
  - **2**: MIT protocol

Response Frame:

Data Field	11-bit ID	8-Byte Data Area
Size	Bit 10~0	Byte 0~7
Description	Motor ID	64-bit MCU unique identifier

6. MIT Communication Protocol Description

The motor communication adopts the CAN 2.0 interface with a default baud rate of 1 Mbps. The baud rate can be modified by switching to the private protocol. The standard frame format is as follows:

Data Field	11-bit ID		8-byte Data Area
Size	Bit 10~8	Bit 7~0	Byte 0~7
Description	Mode type	ID	

- Supported Control Modes:
- **MIT Mode**: Provides five motion control parameters to the motor.
- **Velocity Mode**: Specifies the target speed for the motor.
- **Position Mode**: Specifies the target position and speed, allowing the motor to run to the designated position at the configured speed.

6.1. Response Command 1: Data Feedback (Motor Status)

Data Field	11-bit ID	8-byte Data Area
Size	bit10~0	Byte0~Byte7
	0FD	7F 00 00 00 00 00 00 00
Description	Host ID	Byte0: Motor CAN ID Byte1~2: Current angle [0~65535], corresponding to (-12.57 rad ~ 12.57 rad) Byte3: High 8 bits Byte4[7-4] (high 4 bits) as low 4 bits: Current speed [0~4096], corresponding to (-20 rad/s ~ 20 rad/s) Byte4[3-0] (low 4 bits) as high 4 bits Byte5: Low 8 bits: Current torque [0~4096], corresponding to (-60 N·m ~ 60 N·m) Byte6[7-6]: Operation mode0: Reset Mode 1: Cali Mode 2: Motor Mode

		Byte6[5]: 1 = Motor fault, 0 = No motor fault Byte6[4]: 1 = Motor warning, 0 = No warning Byte6[3-0] (low 4 bits) as high 4 bits Byte7: Low 8 bits: Winding temperature = Temperature(°C) × 10, Unit: °C

6.2. Response Command 2: MCU Identification

Data Field	11-bit ID	8-byte Data Area
Size	Bit 10~0	Byte 0~7
Description	Motor ID	64-bit MCU unique identifier

6.3. Command 1: Enable Motor Operation

Data Field	11-bit ID	8-byte Data Area
Size	Bit 10~0	Byte 0~7
	07F	FF FF FF FF FF FF FF FC
Description	Target motor CAN ID	FF FF FF FF FF FF FF FC

Response: Response Command 1

6.4. Command 2: Stop Motor Operation

Data Field	11-bit ID	8-byte Data Area
Size	Bit 10~0	Byte 0~7
	07F	FF FF FF FF FF FF FF FD
Description	Target motor CAN ID	FF FF FF FF FF FF FF FD

Response: Response Command 1

6.5. Command 3: MIT Dynamic Parameters

Data Field	11-bit ID	8-byte Data Area
Size	bit10~0	Byte0~Byte7
	07F	00 00 00 00 00 00 00 00
Description	Target motor CAN ID	Byte0~1: Target angle [0~65535], corresponding to (-12.57 rad ~ 12.57 rad) Byte2: High 8 bits Byte3[7-4] (high 4 bits) as low 4 bits: Target speed [0~4096], corresponding to (-20 rad/s ~ 20 rad/s)

		Byte3[3-0] (low 4 bits) as high 4 bits Byte4: Low 8 bits: Kp [0~4096], corresponding to 0 ~ 5000 Byte5: High 8 bits Byte6[7-4] (high 4 bits) as low 4 bits: Kd [0~4096], corresponding to 0 ~ 100 Byte6[3-0] (low 4 bits) as high 4 bits Byte7: Low 8 bits: Target torque [0~4096], corresponding to (-60 N·m ~ 60 N·m)

Response: Response Command 1

6.6. Command 4: Set Zero Position (Non-Position Mode)

Data Field	11-bit ID	8-byte Data Area
Size	Bit 10~0	Byte 0~7
	07F	FF FF FF FF FF FF FF FE
Description	Target motor CAN ID	FF FF FF FF FF FF FF FE

Response: Response Command 1

6.7. Command 5: Clear Errors & Read Fault Status

Data Field	11-bit ID	8-byte Data Area
Size	bit10~0	Byte0~Byte7
	07F	FF FF FF FF FF FF FF FB
Description	Target motor CAN ID	FF FF FF FF FF FF F_CMD FB F_CMD:- 0xFF → clearing the current abnormality; Any other value → he error code will be returned in the BYTE1 field of the response message.

Response (Fault Clear): Response Command 1

6.8. Fault Status Response:

Data Field	11-bit ID	8-byte Data Area
Size	bit10~0	Byte0~Byte7
	0FD	7F 00 00 00 00 00 00 00
Description	Target motor CAN ID	Byte0: Motor CAN ID Byte1~4: Fault value (Non-zero: fault; 0: normal) Bit16: Motor current fault: Phase A sampling overcurrent

		Bit14: Motor stall and overload algorithm protection Bit9: Position initialization fault Bit8: Hardware identification fault Bit7: Encoder uncalibrated Bit5: Motor current fault: Phase C sampling overcurrent Bit4: Motor current fault: Phase B sampling overcurrent Bit3: Overvoltage fault Bit2: Undervoltage fault Bit1: Driver chip fault Bit0: Motor overtemperature fault, default: 145 ℃

6.9. Command 6: Set Operation Mode

Data Field	11-bit ID	8-byte Data Area
Size	bit10~0	Byte0~Byte7
	07F	FF FF FF FF FF FF 00 FC
Description	Target motor CAN ID	FF FF FF FF FF FF F_CMD FC F_CMD: Mode type 0: MIT mode (default) 1: Position mode 2: Velocity mode

Response: Response Command 1

6.10. Command 7: Modify Motor CAN ID

Data Field	11-bit ID	8-byte Data Area
Size	bit10~0	Byte0~Byte7
	07F	FF FF FF FF FF FF 01 FA
Description	Target motor CAN ID	FF FF FF FF FF FF F_CMD FA F_CMD: Target motor CAN ID

Response: Response Command 2

6.11. Command 8: Change Communication Protocol (Takes Effect After Power Cycle)

Data Field	11-bit ID	8-byte Data Area
Size	bit10~0	Byte0~Byte7
	07F	FF FF FF FF FF FF 00 FD
Description	Target motor CAN ID	FF FF FF FF FF FF F_CMD FD F_CMD: Protocol type 0: Private protocol (default) 1: CANopen 2: MIT protocol

Response: Response Command 2

6.12. Command 9: Modify Host CAN ID

Data Field	11-bit ID	8-byte Data Area
Size	bit10~0	Byte0~Byte7
	07F	FF FF FF FF FF FF FD 01
Description	Target motor CAN ID	FF FF FF FF FF FF F_CMD 01 F_CMD: Host CAN ID

Response: Response Command 2

6.13. Command 10: Position Mode Control Command

Data Field	11-bit ID		8-byte Data Area
Size	bit10~8	bit7~0	Byte0~Byte7
	1	7F	00 00 A0 40 00 00 A0 40(Position: 5 rad, Speed: 5 rad/s)
Description	1	Target motor CAN ID	Byte 0~3: Target position (rad, 32-bit float) Byte 4~7: Target speed (rad/s, 32-bit float)

Response: Response Command 1

6.14. Command 11: Velocity Mode Control Command

Data Field	11-bit ID		8-byte Data Area
Size	bit10~8	bit7~0	Byte0~Byte7
	2	7F	00 00 A0 40 00 00 00 40 (Speed: 5 rad/s, Current: 2 A)
Description	2	Target motor CAN ID	Byte 0~3: Target speed (rad/s, 32-bit float) Byte 4~7: Current limit in speed/position mode (A, 32-bit float)

Response: Response Command 1

6.15. Command 12:Save motor data (requires firmware update to 0.3.1.41)

Data Field	11-bit ID	8-byte Data Area
Size	Bit 10~0	Byte0~Byte7
	07F	FF FF FF FF FF FF FF F8
Description	Target motor CANID	FF FF FF FF FF FF FF F8

Response frame: Response command 2

6.16. Command 13: Motor actively reports (requires firmware update to 0.3.1.41)

Data Field	11-bit ID	8-byte Data Area
Size	Bit 10~0	Byte0~Byte7
	07F	FF FF FF FF FF FF 00 F9
Description	Target motor CANID	FF FF FF FF FF FF F_CMD F9 The F_CMD byte indicates the motor protocol type. 0 indicates no reporting (default). 1 is for reporting

Response frame: Response command 1

6.17. Command 14: Read motor parameters (requires firmware update to 0.3.1.41)

Data Field	11-bit ID		8Byte 数据区
Size	bit10~8	bit7~0	Byte0~Byte7
	3	7F	05 70 00 00 00 00 00 00
Description	3	Target motor	Byte0~1: index(See the read/write parameter

		CANID	table below for details.) Byte2~3: 00 Byte4~7: 00 The above data is in low byte first, high byte last.

Response frame:

Data Field	11-bit ID		8Byte 数据区
Size	bit10~8	bit7~0	Byte0~Byte7
	3	7F	05 70 00 00 00 00 00 00
Description	3	Target motor CANID	Byte0~1: index See the read/write parameter table for details (same as the private protocol parameter table). Byte2~3: 00 Byte4~7: 00 For parameter data, data of 1 byte or more is formatted with the least significant byte first and the most significant byte last.

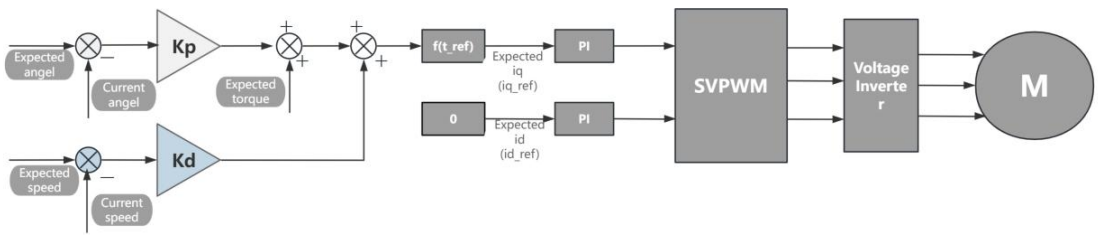
6.18. Command 15: Write motor parameters (requires firmware update to 0.3.1.41)

Data Field	11-bit ID		8Byte 数据区
Size	bit10~8	bit7~0	Byte0~Byte7
	4	7F	29 70 00 00 00 00 00 00
Description	4	Target motor CANID	Byte0~1: index See the read/write parameter table for details (same as the private protocol parameter table). Byte2~3: 00 Byte4~7: 00 Parameter data: The above data is in low byte first, high byte last.

Response frame:

Data Field	11-bit ID		8Byte 数据区
Size	bit10~8	bit7~0	Byte0~Byte7
	4	7F	29 70 00 00 00 00 00 00
Description	4	Target motor CANID	Byte0~1: index See the read/write parameter table for details (same as the private protocol parameter table). Byte2~3: 00 Byte4~7: 00 Parameter data: The above data is in low byte first, high byte last.

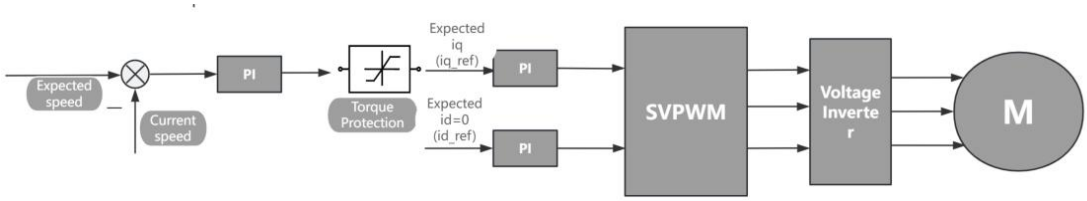
6.19. Motion Control Mode



The motor defaults to Motion Control Mode upon power-up.

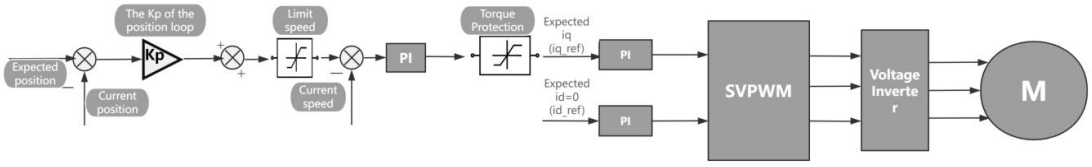
1. Send the *Motor Enable Command* (Command 1).
2. Send the *Motion Control Command* (Command 3) to activate dynamic parameter control.
3. Send the *Motor Stop Command* (Command 2) to halt operation when needed.

6.20. Velocity Mode



1. Configure the motor's operation mode by sending *Set Operation Mode Command* (Command 6) with **Mode = 2 (Velocity Mode)**.
2. Send the *Motor Enable Command* (Command 1) to activate the motor.
3. Send the *Velocity Mode Control Command* (Command 11) to set the **maximum current (absolute value)** and **target speed**.
4. To stop, send the *Motor Stop Command* (Command 2).

6.21. Position Mode (CSP - Cyclic Synchronous Position)



1. Configure the motor's operation mode by sending *Set Operation Mode Command* (Command 6) with **Mode = 1 (Position Mode)**.

2. Send the *Motor Enable Command* (Command 1) to activate the motor.

3. Send the *Position Mode Control Command* (Command 10) to set the **maximum speed (absolute value)** and **target position**.

4. To stop, send the *Motor Stop Command* (Command 2).

7. Version History

Version Number	Description	Date
1.0	Initial Release	November 25, 2025